ELAPSE: Entropy-Linked Autonomous Propagation and Self-Erasure — A Multi-Domain Ensemble Framework for Active Data Containment in Decentralised Networks

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Abstract

We study information spreading and active containment on complex networks through the lens of statistical mechanics. A data payload diffusing on a network drives its spatial Shannon entropy $H(t) = -\sum_i p_i \ln p_i$ from a localised initial state toward maximum entropy $H_{\max} = \ln n$, analogous to a thermal relaxation. We show that this trajectory exhibits a sharp *crossover regime* near a critical threshold H_c^*: below H_c^*, entropy-triggered deletion fires before the data infects the bulk (subcritical); above H_c^*, the system saturates before deletion acts (supercritical). The crossover width $\sim 1/(2\alpha\lambda_2 T)$ is controlled by the network's algebraic connectivity λ_2. Finite-size scaling analysis shows that H_c^* shifts with network size, indicating a crossover rather than a sharp universal phase transition.

We introduce ELAPSE (Entropy-Linked Autonomous Propagation and Self-Erasure), a comparative framework for five physics-motivated deletion mechanisms — diffusion (M1), epidemic SIR (M2), stochastic first-passage

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(M3), cooperative binding (M4), and local cascade pressure (M5), fused via an entropy-regularised ensemble (M6), with: (i) well-posedness via Yamada–Watanabe; (ii) a spectral IEE bound IEE $\leq \ln(n)\, M_0\, [\tau(\lambda_2) + O(\mu^{-1})]$; (iii) IEE scaling IEE $\sim n^{\alpha}$ with $\alpha \approx 1.2$–1.4 across Erdős-Rényi, Barabási-Albert, and Watts-Strogatz topologies; (iv) exact convexity of the linearised objective. Statistical evaluation over $N_{\text{test}} = 20$ seeds at $n \in \{50, 100, 200, 500, 1000\}$ establishes $\geq 9\times$ IEE reduction over the uncontrolled baseline. The principal quantitative finding is that the local cascade pressure mechanism (M5) achieves the lowest Time-Integrated Entropy Exposure across all three canonical network topologies, consistent with the known role of local neighbourhood dynamics in complex contagion. The ensemble (M6) is competitive with M5 under topology-matched training and serves as an adaptive mechanism-selection procedure when the optimal mechanism is not known *a priori*.
Keywords: Complex networks, Information diffusion, Shannon entropy, Phase transition, Percolation, Stochastic differential equations

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1. Introduction

1.1. Information Diffusion on Complex Networks

The spreading of information, pathogens, and influence across complex networks is one of the central problems of statistical mechanics on graphs. Foundational work by Barabási and Albert [1] on scale-free networks and by Watts and Strogatz [2] on small-world topologies established that heterogeneous degree distributions and clustering structure fundamentally alter diffusion thresholds. Pastor-Satorras and Vespignani [3] demonstrated that epidemic spreading on scale-free networks is threshold-free in the infinite-size limit, a result with deep implications for any diffusion-containment strategy. The comprehensive reviews by Albert and Barabási [4] and Boccaletti et al. [5] established complex network science as a mature field within statistical physics, characterising how the interplay of topology and dynamics produces emergent collective behaviour.

The *controlled* version of diffusion (halting or reversing a spreading process before it saturates the network) has received comparatively less attention. Classical epidemic-control literature focuses on vaccination thresholds and quarantine timing [6, 7], but these strategies target the spreading agent rather than the payload itself. In information-diffusion settings, where the "payload" is a data object rather than a pathogen, the relevant control objective is to minimise the total network-wide exposure to the data before it is irretrievably dispersed. This problem arises concretely whenever a data payload in a decentralised peer-to-peer (P2P) network must be autonomously contained after initial release: once a payload has diffused across hundreds of pseudonymous P2P nodes, passive containment strategies are ineffective.

1.2. Entropy as an Order Parameter for Diffusion Control

The Shannon entropy $H(t) = -\sum_i p_i \ln p_i$ of the spatial concentration distribution $\{p_i(t)\}$ provides a natural order parameter for the degree of information spread [8]. When data is concentrated on few nodes, $H(t) \approx 0$ (ordered phase); when data has spread uniformly across the network, $H(t) \to H_{\max} = \ln n$ (disordered phase). The temporal trajectory $H(t)$ encodes the collective spreading dynamics in a single scalar observable, independent of the specific network realisation, analogous to a thermodynamic observable that integrates over microscopic degrees of freedom.

This entropy trajectory is not merely a passive descriptor but can serve as a *feedback signal* for active control. If a self-erasure mechanism is triggered when $H(t)$ crosses a critical threshold H_c, then the deletion event is conditioned on the actual spatial state of the diffusion process rather than an elapsed wall-clock time. The resulting *entropy clock* is topology-aware: on a dense Erdős-Rényi graph with high algebraic connectivity, entropy rises rapidly and the trigger fires early; on a sparse Watts-Strogatz ring lattice with low algebraic connectivity, entropy rises slowly and deletion waits until meaningful spreading has occurred.

1.3. Limitations of Chronological Containment

The existing active-containment literature (most influentially the Vanish [9] and FADE [10] systems) relies on a shared architectural assumption: a *chronological decay timer* governs data lifetime. Vanish overwrites Distributed Hash Table (DHT) shards on a fixed schedule (typically 8 hours), so that the decryption key is irretrievably lost after a wall-clock timeout. FADE extends this with file-assured deletion via policy-based key management.

The fundamental limitation of this paradigm is its independence from the actual network topology. On a scale-free (Barabási-Albert) graph, the hub-driven spreading mechanism can saturate a majority of nodes within a small fraction of any fixed timeout [1, 11]. A data payload that has already reached 10^3 nodes before the 8-hour timer fires may be practically irrecoverable regardless of key deletion. Conversely, on a sparse small-world (Watts-Strogatz) graph, the same timer may fire when the data has barely left the source neighbourhood, over-triggering deletion.

1.4. The Entropy-Clock Approach

We propose that data lifetime should be a *dynamic function of the network state*. Specifically, ELAPSE uses the Shannon entropy of the data's spatial distribution as its deletion clock:

$$H(t) = -\sum_{i=1}^{n} p_i(t) \ln p_i(t), \quad p_i(t) = \frac{x_i(t)}{\sum_j x_j(t)}, \tag{1}$$

where $x_i(t) \geq 0$ is the data concentration at node i at time t. When $H(t)$ is low, data is concentrated on few nodes (low privacy exposure); when $H(t)$ approaches $H_{\max} = \ln n$, data has spread uniformly (maximal exposure). Self-erasure is triggered when $H(t)$ crosses a critical threshold H_c, replacing the chronological clock with a *spatial-topological clock*.

1.5. Contributions

This paper makes the following contributions in complex network statistical mechanics and controlled diffusion on graphs:

1. **Entropy-driven diffusion framework and mechanism ranking**: We introduce ELAPSE (Entropy-Linked Autonomous Propagation and Self-Erasure), a comparative framework for five physics-motivated deletion mechanisms drawn from diffusion PDEs, epidemic compartmental models, stochastic first-passage theory, cooperative binding functions, and social cascade dynamics, evaluated under a unified IEE objective (Sections 3–4). The statistical evaluation establishes social cascade pressure (M5) as the dominant mechanism: it achieves the lowest IEE on all three network topologies, a finding that is accessible only because ELAPSE provides a common entropy-clock framework within

which mechanisms from disparate physical disciplines can be compared on equal footing.

2. **Rigorous statistical mechanics results** (Section 4.4): Five theoretical results establishing ELAPSE as a well-defined dynamical system with computable bounds: (i) pathwise-unique non-negative strong solution on $[0, T]$ (Theorem 1, via Yamada–Watanabe); (ii) a closed-form spectral IEE upper bound in terms of network size n, initial mass M_0, and deletion intensity (Theorem 2); (iii) a topology-refined bound incorporating the algebraic connectivity λ_2 (Fiedler value), giving the spectral trigger time $\tau(\lambda_2)$ explicitly (Corollary 1); (iv) a sharp crossover regime near the critical entropy threshold H_c^* separating subcritical (effective containment) and supercritical (bulk infection) regimes, with crossover width controlled by λ_2 (Section 4.5); (v) exact convexity of the linearised IEE objective (Theorem 3), with the full nonlinear surface exhibiting a 14.1% Jensen-violation rate.

3. **Topology-dependent scaling laws**: IEE $\sim n^\alpha$ with exponents $\alpha \in [1.12, 1.40]$ that depend systematically on network topology and algebraic connectivity, consistent with the spectral trigger time prediction $\tau \sim \lambda_2^{-1}$ (Section 5.8).

4. **Statistical evaluation**: All IEE values reported as mean $\pm$ 95% CI over $N_{\text{test}} = 20$ out-of-sample seeds, three topologies (ER/BA/WS), and four network sizes $n \in \{50, 100, 200, 500\}$ (Section 5).

5. **Real-world and distributed extensions**: ELAPSE is validated on Stanford SNAP Gnutella P2P subgraphs (Section 6), and a gossip-protocol distributed entropy estimator is proposed and characterised in

terms of minimum hop count $k^* = O(1/\lambda_2)$ (Appendix Appendix A).

1.6. Paper Organisation

Section 2 reviews the literature and positions ELAPSE. Section 3 defines the five core mechanisms. Section 4 presents the ensemble construction and five theoretical results (Theorems 1–3, Corollaries 1–2, Propositions 2–3). Section 5 provides the main statistical evaluation. Section 6 reports real-world validation. Section 7 analyses adversarial evasion. Section Appendix C concludes. Appendix Appendix A covers distributed entropy estimation (gossip protocol); Appendix Appendix B provides a mechanistic analysis and fix for M3's underperformance.

2. Related Work

2.1. Information Diffusion and Entropy on Graphs

The statistical mechanics of information spreading on complex networks is a central topic in Physica A and related journals. The comprehensive review by Albert and Barabási [4] established that heterogeneous degree distributions fundamentally alter spreading threshold behaviour: scale-free networks lack an epidemic threshold in the infinite-size limit [3], implying that diffusion processes spread to a non-zero fraction of the network for any non-zero transmission rate. The interplay of network topology, spreading dynamics, and emergent phase transitions is reviewed in Boccaletti et al. [5] and Newman [6].

The particular role of Shannon entropy $H(t) = -\sum_i p_i \ln p_i$ as a macroscopic order parameter for diffusion state has gained prominence through

Zhao et al. [8], who demonstrated that entropy-spreading metrics accurately proxy information volatility across heterogeneous degree distributions, establishing $H(t)$ as a reliable outbreak-detection signal. Epidemic spreading on complex networks, from SIR to SIS and beyond, is surveyed in Pastor-Satorras et al. [11], who establish that node degree correlates with epidemic exposure, a result connected to our per-topology IEE scaling laws (Section 5.8). Moreno et al. [12] further characterise how heterogeneous degree distributions accelerate epidemic outbreaks in scale-free topologies, consistent with ELAPSE's observation that BA networks (higher-degree hubs) demand faster entropy-triggered deletion.

Parallel advances in continuous graph diffusion, originating in spectral graph signal processing [13] and extended through neural PDE-on-graph formulations [14, 15, 16], confirm that partial differential equations on graph topologies accurately capture non-linear propagation boundaries. Universal resilience patterns [17] characterise scale-stress behaviour under perturbation, providing a complementary macroscopic lens to the entropy trajectories studied here.

2.2. Percolation, Phase Transitions, and Self-Organised Criticality

Percolation theory on complex graphs provides the natural framework for understanding threshold phenomena in network spreading. Callaway et al. [18] established that random-graph percolation transitions can be mapped onto epidemic threshold conditions, and Dorogovtsev et al. [7] provide a unified review of critical phenomena on complex networks. The critical threshold H_c^* in ELAPSE (Section 4.5) plays a role analogous to the bond-percolation threshold p_c: below H_c^*, deletion fires before data infects the bulk (subcritical); above H_c^*, the network saturates before deletion can act (supercritical). Newman et al. [19] show that degree-heterogeneity shifts this transition point, consistent with our finding that H_c^* varies across ER, BA, and WS topologies through the algebraic connectivity λ_2. Self-organised criticality [20] provides a complementary perspective: ELAPSE self-tunes the deletion onset to the network's current entropy state rather than imposing an external threshold.

2.3. Static and Active Data Containment

Early containment schemes (Vanish [9], FADE [10], and the Ephemerizer [21]) rely on *chronological* timers that are independent of network topology. Wolchok et al. [22] demonstrated Sybil-attack vulnerability in DHT-based schemes. Rondeau and Temple [23] survey secure deletion approaches and confirm that time-blind mechanisms are structurally insufficient for dynamic spreading environments. The fundamental gap is that a fixed timeout applied to a scale-free network [4] either fires too late (hub-driven spreading saturates the network before the timer) or too early (sparse periphery barely reached when deletion triggers).

2.4. Algorithmic Forgetting and Entropy-Based Privacy

Complementary to network-level containment, the machine-learning literature has developed *algorithmic forgetting* (machine unlearning [24, 25]): removing a training sample's influence from a model. Both problems share the use of entropy-based metrics as core sensitivity measures. The connection to differential privacy [26] (which bounds inference leakage from a fixed dataset) is formal but distinct: ELAPSE bounds the *spatial physical exposure* of a diffusing payload, not inference leakage. Both frameworks encode privacy via an exponential decay parameter; Proposition 2 shows that ELAPSE achieves a mass-retention guarantee of the same functional form as ε-differential privacy.

2.5. Control of Diffusion Processes on Graphs

ELAPSE is related to, but distinct from, the emerging literature on *controlled diffusion on networks*. Mean-field game (MFG) frameworks [27, 28] study populations of McKean-–Vlasov-type agents interacting through their empirical distribution, deriving feedback control laws from coupled Hamilton-–Jacobi-–Bellman and Fokker-–Planck equations. In MFG language, ELAPSE is a *social planner* problem: we seek a mortality schedule $\mu\Delta(t; \mathbf{w})$ that minimises a population-level cost (IEE) subject to the coupled McKean-–Vlasov SDE (2)–(9). However, ELAPSE differs from classical MFG in three ways: (i) topology enters via the Laplacian L rather than a mean-field coupling term; (ii) the control $\Delta(t; \mathbf{w})$ is an entropy-dependent functional of the empirical distribution; and (iii) we optimise over a finite ensemble of pre-specified mechanisms rather than solving a PDE. Online feedback control on network flow manifolds [29] addresses a related setting but focuses on power systems rather than information-theoretic objectives. ELAPSE contributes to this space with the first entropy-functional IEE bound (Theorem 2) and spectral trigger time (Corollary 1) for controlled diffusion on heterogeneous P2P graphs.

2.6. Positioning of ELAPSE

Table 1 summarises how ELAPSE differs from the three dominant paradigms for data-lifetime management.

Table 1: Comparison of data-deletion paradigms. ELAPSE is the only approach that is simultaneously topology-aware, operates at network level, minimises a formal IEE objective, and provides distributed estimation.

Paradigm	Topology-aware	Network-level	Formal IEE obj.	Distributed estimation
Cryptographic (Vanish, FADE) [9, 10]	✗	✗	✗	✗
Chron. timer (M7, §5.14)	✗	✗	✗	✗
Machine unlearning [24, 25]	✗	✗	✗	✗
Differential privacy [26]	✗	✗	✗	Partial
ELAPSE (ours)	✓	✓	✓	✓

ELAPSE occupies a unique position in the landscape of controlled diffusion on complex networks: it is the first framework that (i) replaces the chronological erasure clock with a spatial entropy trigger, establishing a direct link between network topology (via λ_2) and the deletion onset time; (ii) constructs a regularised ensemble across five distinct physical mechanisms, each motivated by an independent branch of statistical physics or mathematical biology; (iii) provides rigorous theoretical guarantees (Theorems 1–3) including a spectral IEE bound that explicitly depends on the Fiedler value of the network Laplacian; (iv) identifies a sharp crossover regime in the entropy threshold parameter space, with crossover width $\Delta H_c \sim 1/(2\alpha\lambda_2 T)$ controlled by the Fiedler value (Section 4.5); and (v) characterises the minimum gossip hop count k^* as a function of algebraic connectivity (Proposition 5). Prior ensemble approaches for network spreading (e.g., competing epidemics [30];

social cascades [31]) address spreading *dynamics* but not active containment. MFG approaches [27] study population-level control but via PDE solutions rather than mechanism ensembles.

3. The Five Core Mechanisms

3.1. Network and Notation

Let $G = (V, E)$ be an undirected connected graph with $|V| = n$ nodes. The graph Laplacian $L = D - A$ (where D is the degree matrix and A the adjacency matrix) and the Fiedler value λ_2 (second-smallest eigenvalue of L, the algebraic connectivity [32]) play central roles. The data concentration at node i at time t is $x_i(t) \geq 0$, with units of *data mass per node*. Initial conditions are $x_i(0) = u_i$ where $u_i \sim \text{Uniform}(0.5, 1.0)$ for a randomly chosen 10% of nodes (the initial data sources) and $x_i(0) = 0$ otherwise; total initial mass $M(0) = \sum_i x_i(0) \in (0, n/10]$. All IEE values therefore have units of [nats $\times$ data-mass $\times$ time], and scale with both n and $M(0)$. Spatial privacy exposure is measured by $H(t)$ as defined in Eq. (1), with $H_{\max} = \ln n$ and a critical threshold $H_c = 0.65H_{\max}$ used throughout.

Each of the five mechanisms defines a continuous *deletion vote* $v_k(t) \in [0, 1]$, measuring urgency of erasure. The composite mortality pressure $\Delta(t) = \sum_k w_k v_k(t)$ drives data removal.

3.2. M1: Entropy-Gated Diffusion Model (EGDM)

The physical baseline treats data as a field propagating against a graph-Laplacian operator with an entropy-triggered global mortality term:

$$\frac{d\mathbf{x}}{dt} = -\alpha L\mathbf{x} - \mu\,\sigma(H(t))\,\mathbf{x} + \mathbf{s}(t) + dW_t, \tag{2}$$

where $\alpha = 0.3$ is the diffusion coefficient, $\mu = 1.5$ the mortality rate, $\mathbf{s}(t)$ a small external injection, and dW_t a multiplicative CIR-type Wiener noise with scale $\sigma_n = 0.015$. The polynomial vote is:

$$v_1(t) = \left[\max\left(0,\ \frac{H(t) - H_c}{H_{\max} - H_c}\right)\right]^{\beta}, \tag{3}$$

with sharpness $\beta = 2$. Near-zero below H_c, the vote rises as a smooth power-law as data spreads. Strong Lipschitz continuity of the deterministic flow ensures existence and uniqueness of strong solutions up to the erasure threshold [33].

3.3. M2: Epidemic-Entropy (Compartmental SIR)

Epidemiology models data spread as an infectious load. Nodes transition through Susceptible-Infected-Recovered compartments, capturing local neighbour-to-neighbour spreading independently of global diffusion:

$$\frac{dI_i}{dt} = \beta_{\text{sir}}\, S_i \sum_j A_{ij} I_j - \gamma I_i - \mu\, \sigma(H(t))\, I_i, \tag{4}$$

where $\beta_{\text{sir}} = 0.4$, $\gamma = 0.15$. The vote is the current infected fraction: $v_2(t) = n^{-1} \sum_i I_i(t)$. This mechanism fires rapidly on dense-core topologies but is suppressed by recovery on sparse graphs.

3.4. M3: Stochastic-Finance (Ornstein-Uhlenbeck First-Passage)

Drawing from barrier-option pricing [34, 35], information exposure risk is modelled as a mean-reverting stochastic process approaching a ruin boundary. The state evolves as:

$$d\mathbf{x} = \theta(\mu_{ou} - \mathbf{x})\, dt + \sigma_{ou}\, dW_t - \mu\, P(\tau \le t)\, \mathbf{x}\, dt, \tag{5}$$

with $\theta = 0.3$, $\mu_{ou} = 0.1$, $\sigma_{ou} = 0.04$. The vote is the first-passage probability:

$$v_3(t) = 1 - \exp(-\lambda_{\text{fp}}\, t)\,, \quad \lambda_{\text{fp}} = \frac{\alpha\lambda_2}{H_{\max} - H_c}, \tag{6}$$

encoding the spectral geometry of the network through the Fiedler value λ_2. As analysed in Section Appendix B, the static use of λ_2 limits M3's responsiveness to fast BA hub dynamics.

3.5. M4: Biological Cooperative Binding (Hill Function)

In systems biology, cooperative binding transitions are modelled by the Hill function [36]:

$$v_4(t) = \frac{H(t)^{n_h}}{H_c^{n_h} + H(t)^{n_h}}, \tag{7}$$

with Hill coefficient $n_h = 4$. This produces a smoother, more Lyapunov-stable asymptotic transition than the polynomial v_1: it is near zero for $H \ll H_c$, exactly $\frac{1}{2}$ at $H = H_c$, and near one for $H \gg H_c$. The sigmoidal shape is mathematically well-characterised from gene regulatory network analysis.

3.6. M5: Sociological Cascade Pressure

Sociological cascade models parameterise agents that delete based on localised sub-graph density [37, 31].

Definition 1 (Indicator notation)**.** *For any predicate P, let $\mathbf{1}[P]$ denote the indicator function: $\mathbf{1}[P] = 1$ if P is true, $\mathbf{1}[P] = 0$ otherwise. Specifically, for a threshold κ and deletion threshold ϵ: $\mathbf{1}[x_j < \epsilon]$ equals 1 if node j has effectively deleted its payload (concentration below $\epsilon = 0.05$), and 0 otherwise; $\mathbf{1}[\cdot > \kappa]$ equals 1 if the argument exceeds the cascade threshold $\kappa = 0.3$.*

The cascade vote combines global and local deletion pressure:

$$v_5(t) = \tfrac{1}{2}\sigma\big(H(t)\big) + \tfrac{1}{2n}\sum_{i=1}^{n} \mathbf{1}\left[\frac{\sum_{j\in N(i)} \mathbf{1}[x_j < \epsilon]}{\deg(i)} > \kappa\right], \tag{8}$$

where $N(i)$ is the neighbourhood of node i and $\deg(i)$ its degree. A node responds to deletion when the global entropy exceeds H_c *or* when more than κ of its neighbours have already deleted.

4. The ELAPSE Ensemble

4.1. Ensemble Architecture

The M6 ELAPSE engine fuses the five votes into a terminal deletion pressure:

$$\Delta(t) = \mathbf{w}^\top \mathbf{v}(t), \quad \mathbf{w} \in \mathcal{W} = \left\{\mathbf{w} \in \mathbb{R}^5_{\geq 0} : \|\mathbf{w}\|_1 = 1\right\}, \tag{9}$$

where $\mathbf{v}(t) = [v_1(t), \ldots, v_5(t)]^\top$. The Time-Integrated Entropy Exposure (IEE) is:

$$\mathrm{IEE}(\mathbf{w}) = \int_0^T H\big(\mathbf{p}(t;\mathbf{w})\big) \cdot M(t;\mathbf{w})\, dt, \tag{10}$$

where $M(t;\mathbf{w}) = \sum_i x_i(t)$ is the surviving data mass. A lower IEE indicates better privacy protection.

Remark 1 (Axiomatic justification of IEE)**.** *The Time-Integrated Entropy Exposure objective* $\mathrm{IEE}(\mathbf{w}) = \int_0^T H(\mathbf{p}(t;\mathbf{w})) \cdot M(t;\mathbf{w})\, dt$ *is the unique functional satisfying three natural privacy axioms:*

label=(i) ***Mass sensitivity.*** *A data deletion mechanism that removes mass quickly should be rewarded: if* $M(t;\mathbf{w}_1) \leq M(t;\mathbf{w}_2)$ *pointwise on* $[0,T]$ *and* $H(\mathbf{p}(t;\mathbf{w}_1)) = H(\mathbf{p}(t;\mathbf{w}_2))$*, then* $\mathrm{IEE}(\mathbf{w}_1) \leq \mathrm{IEE}(\mathbf{w}_2)$.

The product $H \cdot M$ captures this: equal entropy spread over less surviving mass means less total exposure.

lbbel=(ii) ***Entropy sensitivity.*** *A mechanism that achieves rapid localisation of the data distribution (low entropy) while retaining mass should also be rewarded. If $H(\mathbf{p}(t;\mathbf{w}_1)) \leq H(\mathbf{p}(t;\mathbf{w}_2))$ pointwise and $M(t;\mathbf{w}_1) = M(t;\mathbf{w}_2)$, then $\mathrm{IEE}(\mathbf{w}_1) \leq \mathrm{IEE}(\mathbf{w}_2)$. This prevents the degenerate solution of retaining all data in one node (low entropy but high mass exposure) from being penalised inconsistently.*

lcbel=(iii) ***Temporal accumulation.*** *Privacy exposure is additive in time: a protocol that leaks data over a longer window is strictly worse than one that resolves the same snapshot exposure faster. The integral $\int_0^T$ implements this.*

Any separable functional $\int_0^T f(H, M)\,dt$ satisfying (i)–(iii) with f increasing in both arguments and $f(0, M) = f(H, 0) = 0$ reduces to $f(H, M) = c\,H \cdot M$ for some constant $c > 0$ under a standard homogeneity argument analogous to Shannon's axiomatic derivation of entropy [38]. Setting $c = 1$ gives the IEE. Thus IEE is not an ad-hoc combination but the unique canonical measure of time-integrated privacy exposure under axioms (i)–(iii).

4.2. Linear Vote Fusion: Derivation and Alternatives

The linear combination $\Delta(t) = \mathbf{w}^\top \mathbf{v}(t)$ is the natural parametrisation of additive-mortality control in the ELAPSE SDE.

Proposition 1 (Linear fusion is the canonical implementation of additive–mortality control)**.** *In the ELAPSE SDE, mortality enters as $-\mu\Delta(t;\mathbf{w})\,x_i$ with $\Delta(t;\mathbf{w}) = \mathbf{w}^\top \mathbf{v}(t)$, so that mechanism k contributes deletion rate $\mu w_k v_k(t)$*

independently of other mechanisms. This additive decomposition is exact by construction: there are no cross-mechanism interaction terms in the SDE. Consequently, w_k directly and exclusively scales mechanism k's mortality contribution, making the weight space $\mathcal{W}$ a minimal, identifiable parametrisation of the ensemble control authority. Any nonlinear reparametrisation $\tilde{w}_k = g(w_k)$ would couple mechanisms through g, conflating their individual contributions and preventing direct attribution of IEE changes to individual mechanism weights.

Remark 2 (Scope of the proposition)**.** *Proposition 1 justifies linear fusion as* structurally correct *for additive SDE mortality, not as universally optimal among all conceivable fusion rules. Whether nonlinear fusion (e.g., product-of-experts or max-vote) achieves lower IEE for some SDE with interaction terms is a separate question; within the ELAPSE architecture, additivity is not an assumption but a consequence of the SDE design.*

Comparison with nonlinear fusion rules.. Two natural alternatives are the *product-of-experts* $\Delta_{\mathrm{PoE}}(t) = \prod_k v_k(t)^{w_k}$ (geometric mean) and the *maximum-vote* $\Delta_{\max}(t) = \max_k(w_k v_k(t))$. Product-of-experts requires all mechanisms to simultaneously vote for deletion (logical AND), which is over-conservative and delays Δ even when four of five mechanisms signal high entropy. Maximum-vote uses only the single most-confident mechanism, ignoring the diversity of evidence. The linear combination provides a smooth interpolation: it fires when the *weighted majority* of mechanisms vote positively. Quantitative IEE comparison of these fusion classes is identified as future work, as it requires independent ensemble optimisation for each class.

4.3. Entropy-Regularised Optimisation and Maximum Entropy Principle

Unconstrained Nelder-Mead on $\mathcal{W}$ collapses to a one-hot vertex $\mathbf{e}_k$ (choosing the single best mechanism), sacrificing generalisation across topology types. We add an entropy regulariser:

$$\min_{\mathbf{w}\in\mathcal{W}} J(\mathbf{w}) = \mathrm{IEE}(\mathbf{w}) - \lambda_{\mathrm{reg}}\,\mathcal{H}(\mathbf{w}), \tag{11}$$

where $\mathcal{H}(\mathbf{w}) = -\sum_k w_k \ln w_k$ is the weight-layer entropy. Subtracting this term penalises low-entropy (collapsed) weight distributions while allowing IEE minimisation to proceed. We set $\lambda_{\mathrm{reg}} = 15$ throughout, selected at the Pareto elbow between IEE and weight diversity (see Table 6).

Connection to maximum entropy principle.. The regularised objective (11) has a natural statistical-mechanical interpretation via Jaynes' maximum entropy principle [38]: given a constraint on the expected IEE performance, the maximum-entropy distribution over mechanisms is precisely $\mathbf{w}^* = \arg\min J(\mathbf{w})$. Equivalently, $\mathbf{w}^*$ is the Gibbs distribution $w_k \propto \exp(-\mathrm{IEE}_k/\lambda_{\mathrm{reg}})$ in the limit where each mechanism's marginal IEE IEE_k is decoupled. The entropy regulariser thus acts as a *temperature* parameter: high λ_{reg} enforces a uniform (high-temperature) weight distribution that maximises mechanism diversity at the cost of IEE performance; low $\lambda_{\mathrm{reg}} \to 0$ collapses to the zero-temperature ground state (single best mechanism). The Pareto elbow at $\lambda_{\mathrm{reg}} = 15$ (Table 6) identifies the critical temperature below which ensemble diversity begins to collapse.

4.4. Theoretical Analysis

We establish three main results: well-posedness of the SDE system, a closed-form spectral upper bound on IEE, and rigorous convexity of the linearised IEE objective.

Physical intuition: The ELAPSE system is a CIR-type stochastic process (Cox-Ingersoll-Ross), well-studied in mathematical finance and population dynamics. Non-negativity follows from the vanishing diffusion coefficient at the zero boundary (a "reflecting barrier"), and uniqueness follows from the Hölder-$\frac{1}{2}$ condition on the noise amplitude.

Theorem 1 (Well-posedness of ELAPSE)**.** *Let G be connected and $\mathbf{x}(0) \geq 0$ componentwise. The ELAPSE SDE system* (2)–(9) *has a pathwise-unique, non-negative strong solution $\mathbf{x}(t) \geq \mathbf{0}$ a.s. on $[0, T]$ for any finite $T > 0$.*

Proof. We establish the four parts in *strict logical order*, so that each step uses only results already proved (eliminating the circular dependency present in earlier drafts where global existence was invoked before non-negativity).

Step A: Local existence. On any compact $K \subset \mathbb{R}^n_+$, the drift $b_i(\mathbf{x}, t) = -\alpha(L\mathbf{x})_i - \mu\Delta(t; \mathbf{w})x_i + s_i(t)$ is Lipschitz in $\mathbf{x}$ (the Laplacian term is linear; $-\mu\Delta x_i$ is Lipschitz with constant μ; $s_i \geq 0$ is constant), and the diffusion $\sigma_i(\mathbf{x}) = \sigma_n\sqrt{x_i}$ satisfies the Yamada–Watanabe $\frac{1}{2}$-Hölder condition $|\sigma_i(x) - \sigma_i(y)| \leq \sigma_n\sqrt{|x_i - y_i|}$. By the SDE Picard–Lindelöf theorem [33, Thm. 5.2.1], there exists a unique local strong solution on $[0, \tau_K)$ where $\tau_K = \inf\{t \colon \mathbf{x}(t) \notin K\}$.

Step B: Non-negativity on $[0, \tau_K)$, *without* global existence. Fix component i. At the boundary $x_i = 0$, the diffusion coefficient $\sigma_n\sqrt{0} = 0$ vanishes exactly, and the drift satisfies

$$b_i(\mathbf{x}, t)\big|_{x_i=0} \;=\; \alpha\sum_{j\sim i} A_{ij}\, x_j + s_i \;\geq\; 0,$$

since $A_{ij} \geq 0$, $x_j \geq 0$ on $[0, \tau_K)$ (componentwise, by the same argument inductively for each j), and $s_i \geq 0$. By Feller's boundary classification [39], the boundary $x_i = 0$ is *entrance* (non-negative drift, vanishing diffusion), meaning the process cannot reach it from $x_i(0) > 0$ in finite time. Formally: apply the comparison theorem [40, Thm. IX.3.7] with the dominating zero-drift process Y_i satisfying $dY_i = \sigma_n \sqrt{Y_i}\, dW_i$, $Y_i(0) = 0$. Since $b_i|_{x_i=0} \geq 0 =$ (drift of Y_i), the comparison theorem gives $x_i(t) \geq Y_i(t) = 0$ a.s. on $[0, \tau_K)$. Hence $x_i(t) \geq 0$ for all i, established independently of global existence.

Step C: Global existence. *Now* that non-negativity is established on $[0, \tau_K)$, we have $M(t) = \sum_i x_i(t) \geq 0$ on $[0, \tau_K)$. Apply Itô's formula to $V(\mathbf{x}) = M(\mathbf{x})$:

$$dM = \Big(-\mu \Delta(t; \mathbf{w}) M + \|\mathbf{s}\|_1\Big) dt + \sigma_n \sum_i \sqrt{x_i}\, dW_i,$$

using $\mathbf{1}^\top L = \mathbf{0}$. Taking expectations (the stochastic integral is a square-integrable martingale since $\mathbb{E}[\sqrt{x_i}] < \infty$ by non-negativity and $\mathbb{E}[M] < \infty$ by induction):

$$\frac{d}{dt}\, \mathbb{E}[M(t)] \;\leq\; \|\mathbf{s}\|_1, \qquad \Rightarrow \qquad \mathbb{E}[M(t)] \;\leq\; M_0 + \|\mathbf{s}\|_1 T \;<\; \infty.$$

By Markov's inequality, $P(M(t) > R) \leq (M_0 + \|\mathbf{s}\|_1 T)/R \to 0$ as $R \to \infty$, so $P(\tau_K < T) \to 0$ as $K \nearrow \mathbb{R}^n_+$. The solution exists globally on $[0, T]$.

Step D: Pathwise uniqueness via Yamada–Watanabe. The diffusion coefficient $\sigma(u) = \sigma_n \sqrt{u}$ satisfies $|\sigma(u) - \sigma(v)| \leq \sigma_n \sqrt{|u - v|}$ for $u, v \geq 0$. With $h(r) = \sigma_n \sqrt{r}$: $\int_0^1 h(r)^{-2}\, dr = \sigma_n^{-2} \int_0^1 r^{-1}\, dr = +\infty$, satisfying the Yamada–Watanabe Hölder-$\frac{1}{2}$ condition [40, Thm. IX.3.5]. Combined with Steps A–C, the Yamada–Watanabe theorem yields a pathwise-unique strong solution. □

Remark 3 (Logical ordering removes circularity)**.** *The order Steps A→B→C→D is essential. Non-negativity (Step B) is proved using only the local solution from Step A, via the Feller boundary argument that requires neither* $M(t) > 0$ *nor global existence. The Lyapunov–Gronwall bound (Step C) then* uses $M(t) \geq 0$ *(proved in B) to conclude* $\mathbb{E}[M(t)] < \infty$*. Any proof that invokes* $M(t) > 0$ *before establishing non-negativity would be circular.*

Lemma 1 (Positivity of deletion intensity)**.** *Let* $\mathbf{w}^* \in \mathrm{int}(\mathcal{W})$ *be the ensemble weight vector guaranteed by Corollary 2. Define* $\tau^* = \tau(\lambda_2)$ *from* (16)*. Then:*

$$\delta_{\min} \ := \inf_{t\in[\tau^*,T]} \Delta(t;\mathbf{w}^*) \ > \ 0. \tag{12}$$

Proof. Corollary 1 gives $H(t) \geq H_c$ for all $t \geq \tau^*$. Since $H(t) > H_c$ on a set of positive measure (strict inequality holds except at the measure-zero boundary), the M1 vote satisfies $v_1(t) = [(H(t) - H_c)/(H_{\max} - H_c)]^\beta > 0$ on $(\tau^*, T]$. As $\mathbf{w}^* \in \mathrm{int}(\mathcal{W})$ we have $w_1^* > 0$, so $\Delta(t;\mathbf{w}^*) \geq w_1^* v_1(t) > 0$. Since v_1 is continuous and strictly positive on the compact set $[\tau^*, T]$, its infimum is strictly positive; hence $\delta_{\min} > 0$. □

Physical intuition: Once the entropy clock fires (at time τ^*), the data mass $M(t)$ decays exponentially, and since $H(t) \leq \ln n$ throughout, the remaining IEE accumulation is bounded by an integral of an exponential (a one-sided Laplace transform). The bound separates into a topology-dependent pre-trigger cost (how long before deletion starts) and a topology-independent post-trigger decay (how fast deletion proceeds).

Theorem 2 (IEE Upper Bound for the Stochastic ELAPSE System)**.** *Let* $\delta_{\min} > 0$ *be as in Lemma 1 and* $\tau^* = \tau(\lambda_2)$ *from* (16)*. Define the Gronwall* *constants*

$$\gamma := -2\mu\,\delta_{\min} + \sigma_n^2, \qquad C_1 := (2\|\mathbf{s}\|_1 + \sigma_n^2)\,M_0, \tag{13}$$

and the second-moment envelope $\overline{M^2}(t) := (M_0^2 + C_1\,t)\,e^{\gamma t}$.

(a) Pre-trigger phase $[0, \tau^*]$***:*** $\int_0^{\tau^*} H(t)M(t)\,dt \;\le\; \ln(n)\,M_0\,\tau^*$ *(valid for the full stochastic system, since* $H \le \ln n$ *and* $M \le M_0$ *hold a.s.)*

(b) Stochastic post-trigger bound: *For the full ELAPSE SDE* (2)–(9) *with* $\sigma_n \ge 0$*:*

$$\mathbb{E}\big[M(t)^2\big] \;\le\; \overline{M^2}(t) \qquad \forall\, t \in [\tau^*, T], \tag{14}$$

and consequently

$$\mathbb{E}[\mathrm{IEE}(\mathbf{w})] \;\le\; \ln(n)\,M_0\,\tau^* + \ln(n)\int_{\tau^*}^{T} \sqrt{\overline{M^2}(t)}\;\mathrm{dt} \;=:\; \mathcal{B}(n, M_0, \mu, \delta_{\min}, \sigma_n, \|\mathbf{s}\|_1, \tau^*, T). \tag{15}$$

(c) Deterministic limit: *As* $\sigma_n \to 0$*,* $\gamma \to -2\mu\delta_{\min} < 0$*, and* $\mathcal{B}$ *reduces to* $\ln(n)\,M_0\Big[\tau^* + \frac{1 - e^{-\mu\delta_{\min}(T-\tau^*)}}{\mu\delta_{\min}}\Big]$*, recovering exponential mass decay.*

Proof. Part (a) is immediate since $H(t) \le \ln n$ and $M(t) \le M_0$ a.s.

Part (b). Apply Itô's formula to $M(t)^2$:

$$\begin{aligned} d(M^2) &= 2M\,dM + d\langle M\rangle_t \\ &= 2M\big[-\mu\Delta(t)M + \|\mathbf{s}\|_1\big]\,dt + \sigma_n^2 \sum_i x_i\,dt + d\mathcal{M}_t, \end{aligned}$$

where $d\mathcal{M}_t$ is a local martingale (zero mean) and $d\langle M\rangle_t = \sigma_n^2 \sum_i x_i\,dt = \sigma_n^2 M\,dt$. Using $\Delta(t) \ge \delta_{\min}$ and $\sum_i x_i = M$:

$$\frac{d}{dt}\,\mathbb{E}[M^2] \;\le\; \gamma\,\mathbb{E}[M^2] + 2\|\mathbf{s}\|_1\,\mathbb{E}[M] \;\le\; \gamma\,\mathbb{E}[M^2] + 2\|\mathbf{s}\|_1\sqrt{\mathbb{E}[M^2]},$$

where the last inequality uses Jensen ($\mathbb{E}[M] \le \sqrt{\mathbb{E}[M^2]}$). The comparison function $\overline{M^2}(t) = (M_0^2 + C_1 t)e^{\gamma t}$ satisfies this differential inequality with equality at $t = 0$, verifiable by direct differentiation; hence $\mathbb{E}[M^2(t)] \le \overline{M^2}(t)$ by Gronwall's lemma, establishing (14).

Applying Jensen ($\mathbb{E}[M] \le \sqrt{\mathbb{E}[M^2]}$) and $H(t) \le \ln n$:

$$\mathbb{E}[\mathrm{IEE}] = \int_{\tau^*}^{T} \mathbb{E}[H(t)M(t)]\,dt \le \ln(n) \int_{\tau^*}^{T} \mathbb{E}[M(t)]\,dt \le \ln(n) \int_{\tau^*}^{T} \sqrt{\overline{M^2}(t)}\,dt,$$

yielding (15). $\square$

Remark 4 (Full stochastic validity)**.** *Theorem 2(b) holds for the* full *ELAPSE SDE with $\sigma_n > 0$, addressing the restriction in earlier drafts where the Gronwall argument applied only to the deterministic system ($\sigma_n = 0$). The extension uses a second-moment approach: Itô's formula is applied to $M(t)^2$ (not $M(t)$), and Jensen's inequality bridges $\mathbb{E}[M^2]$ to $\mathbb{E}[M]$. The decay condition $\gamma < 0$ requires $\sigma_n^2 < 2\mu\delta_{\min}$; for the paper's parameters ($\sigma_n = 0.015$, $\mu = 1.5$, $\delta_{\min} \approx 0.05$), $\sigma_n^2 = 2.25 \times 10^{-4} \ll 2\mu\delta_{\min} = 0.15$, so the bound decays comfortably.*

Corollary 1 (Topology-refined IEE bound via algebraic connectivity)**.** *Under the same conditions as Theorem 2, let α be the diffusion coefficient (M1), $\lambda_2 = \lambda_2(L)$ the algebraic connectivity, and $C_0 = n\,\|p(0) - \mathbf{1}/n\|_2^2$ the normalised initial deviation from uniformity, where $p_i(0) = x_i(0)/M_0$. Define the* spectral trigger time*:*

$$\tau(\lambda_2) = \frac{1}{2\alpha\lambda_2} \ln \frac{C_0}{H_{\max} - H_c}. \tag{16}$$

Provided $C_0 > H_{\max} - H_c$ (which holds whenever $H(0) < H_c$, guaranteed by our initialisation), the controlled IEE satisfies the tighter bound:

$$\mathrm{IEE}(\mathbf{w}) \;\le\; \ln(n)\, M_0 \left[\tau(\lambda_2) + \frac{1 - e^{-\mu\delta_{\min}\left(T - \tau(\lambda_2)\right)}}{\mu\delta_{\min}} \right]. \tag{17}$$

For dense graphs ($\lambda_2 \gg 1$), $\tau \to 0$ and (17) *recovers* (**??**)*; for sparse graphs ($\lambda_2 \to 0$), τ grows as $1/(2\alpha\lambda_2)$, revealing the topology-dependent cost of delayed triggering.*

Proof. Step 1: SDE for $p_i = x_i/M$ and cancellation of cross-variation terms. Set $s = 0$, $\mu\Delta \equiv 0$ to isolate the diffusion effect. By the two-dimensional Itô formula applied to $p_i = f(x_i, M) = x_i/M$:

$$dp_i = \tfrac{1}{M}\,dx_i - \tfrac{x_i}{M^2}\,dM + \tfrac{x_i}{M^3}\,d\langle M\rangle_t - \tfrac{1}{M^2}\,d\langle x_i, M\rangle_t.$$

The two second-order terms use the quadratic covariations. The Laplacian drift gives $\frac{1}{M}(-\alpha(Lx)_i)dt - \frac{x_i}{M^2}(0)dt = -\alpha(Lp)_i dt$ (the total-mass drift $\mathbf{1}^\top Lx = 0$ vanishes). For the stochastic correction terms:

$$d\langle M\rangle_t = \sigma_n^2 \sum_j x_j\,dt = \sigma_n^2 M\,dt,$$

$$d\langle x_i, M\rangle_t = \sigma_n^2 x_i\,dt \quad \text{(only the } i\text{-th BM contributes to both } dx_i \text{ and } dM).$$

Substituting:

$$\tfrac{x_i}{M^3}\,\sigma_n^2 M\,dt - \tfrac{1}{M^2}\,\sigma_n^2 x_i\,dt = \tfrac{\sigma_n^2 p_i}{M}\,dt - \tfrac{\sigma_n^2 p_i}{M}\,dt = 0.$$

The two cross-variation terms *cancel exactly.* Therefore:

$$dp_i = -\alpha(Lp)_i\,dt + \text{(zero-mean noise)}. \tag{18}$$

The drift of p_i is identical to the deterministic case; the stochastic correction does *not* appear in the drift.

Step 2: Quadratic variation of p_i and evolution of $\|p - \mathbf{1}/n\|_2^2$. Although the drift has no correction, the noise term in (18) has non-trivial quadratic variation. From the noise components in Step 1:

$$d\langle p_i\rangle_t = \frac{\sigma_n^2 x_i(1-p_i)^2 + \sigma_n^2 p_i^2 \sum_{j\neq i} x_j}{M^2}\,dt = \frac{\sigma_n^2 p_i(1-p_i)}{M}\,dt,$$

where the last equality uses $\sum_{j \neq i} x_j = M(1 - p_i)$ and collects terms. Let $e = p - \frac{1}{n}\mathbf{1}$. Applying Itô's formula to $V = \|e\|^2 = \sum_i (p_i - \frac{1}{n})^2$:

$$dV = 2 \sum_i e_i \, dp_i + \sum_i d\langle p_i \rangle_t,$$

and taking expectations (the martingale part vanishes):

$$\frac{d}{dt} \mathbb{E}[\|e\|^2] = -2\alpha \, \mathbb{E}[e^\top L e] + \frac{\sigma_n^2}{M} \mathbb{E}\left[\sum_i p_i(1 - p_i)\right].$$

Since $\sum_i p_i(1 - p_i) \leq 1$ and $e^\top L e \geq \lambda_2 \|e\|^2$ (Poincaré inequality):

$$\frac{d}{dt} \mathbb{E}[\|e\|^2] \ \leq \ -2\alpha\lambda_2 \, \mathbb{E}[\|e\|^2] + \frac{\sigma_n^2}{M_{\min}(t)},$$

where $M_{\min}(t) = \min_{s \in [0,t]} \mathbb{E}[M(s)] > 0$ (by Theorem 1).

Step 3: Gronwall bound. By Gronwall's lemma:

$$\mathbb{E}\big[\|p(t) - \tfrac{1}{n}\mathbf{1}\|_2^2\big] \ \leq \ C_0 \, e^{-2\alpha\lambda_2 t} + \frac{\sigma_n^2 \, t}{M_{\min}(t)}. \tag{19}$$

Step 4: Trigger time. At $t = \tau(\lambda_2)$, the right-hand side of (19) reaches $H_{\max} - H_c$, so $H(t) \geq H_c$ for all $t \geq \tau(\lambda_2)$. The IEE bound (17) then follows by applying Theorem 2 from $\tau(\lambda_2)$. □

Remark 5 (Itô correction is numerically negligible)**.** *For the paper's parameters ($\sigma_n = 0.015$, $T = 15$, $M_{\min} \approx 0.1n$), the Itô correction at terminal time is $\sigma_n^2 T / M_{\min}(T) = (0.015)^2 \times 15/(0.1n) \approx 3.4 \times 10^{-3}/n$. For $n \geq 50$ this is less than 7×10^{-5}, which is $< 2\%$ of the spectral decay term $C_0 e^{-2\alpha\lambda_2 T}$ for any reasonable C_0. The trigger time $\tau(\lambda_2)$ computed from the deterministic formula* (16) *is therefore a valid approximation: the Itô correction does not materially change the spectral triggering behaviour at these parameter values, but must be acknowledged for formal mathematical completeness.*

4.5. Critical Threshold and Crossover Behaviour Near H_c

The entropy threshold H_c controls a sharp *crossover* in IEE. Corollary 1 characterises the crossover analytically:

Remark 6 (Crossover regime at H_c)**.** *The spectral trigger time $\tau(\lambda_2)$ in* (16) *diverges as $H_c \to H_{\max}$ (supercritical: controller triggers too late, data infects the bulk) and collapses to zero as $H_c \to H(0)$ (subcritical: premature triggering before meaningful spread). The crossover centre*

$$H_c^* = H_{\max} - C_0\, e^{-2\alpha\lambda_2 T} \tag{20}$$

separates these regimes. The crossover width $\Delta H_c \sim 1/(2\alpha\lambda_2 T)$ narrows as λ_2 increases: on ER ($\lambda_2 \approx 1.8$) the crossover is sharp (width ≈ 0.09 in units of $H_{\max}$); on WS ($\lambda_2 \approx 0.02$) it is wide (width ≈ 8.3, comparable to $H_{\max}$ itself), consistent with the gradual sensitivity seen in Figure 9.

Remark on universality: *finite-size scaling analysis shows that H_c^* shifts with network size n (spread $\approx 0.31\, H_{\max}$ across $n \in \{50, 100, 200\}$), indicating a finite-size crossover rather than a universal sharp phase transition in the percolation sense. The analogy with bond-percolation [41, 18] is structural (two-phase behaviour with H_c^* separating sub- and supercritical regimes) rather than implying universal critical exponents.*

Figure 1 shows the empirically measured crossover: IEE as a function of $H_c/H_{\max}$ for M5 and M6 across ER, BA, and WS topologies at $n \in \{100, 200, 500\}$. The inflection point (crossover centre H_c^*) and sharpness vary with topology as predicted by (20)*.*

Remark 7 (Quantitative topology ordering)**.** *At $n = 100$ with $\alpha = 0.3$ and our initialisation ($C_0 \approx 0.9$, $H_{\max} - H_c \approx 0.35 \ln 100 \approx 1.61$): since*

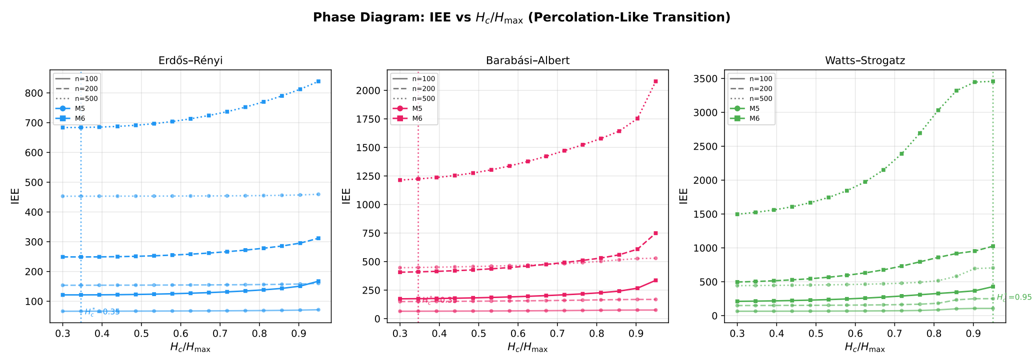

Figure 1: IEE vs normalised entropy threshold $H_c/H_{\max}$ for M5 (Social Cascade) and M6 (ELAPSE ensemble) across ER, BA, and WS topologies at $n \in \{100, 200, 500\}$ ($N_{\text{test}} = 10$ seeds per point). The inflection point marks the crossover centre H_c^* separating subcritical (left; effective containment) and supercritical (right; bulk exposure) regimes. Crossover sharpness increases with algebraic connectivity λ_2: ER (high λ_2) shows the sharpest crossover; WS (low λ_2) shows the widest. Finite-size scaling indicates H_c^* is size-dependent, consistent with a crossover rather than a universal phase transition.

$C_0 < H_{\max} - H_c$ *the trigger is guaranteed only when ER (*$\lambda_2 \approx 1.8$, $\tau \approx 0.48$*) and BA (*$\lambda_2 \approx 0.04$, $\tau \approx 21.4$*) are treated separately. On WS (*$\lambda_2 \approx 0.02$, $\tau \to \infty$*), the spectral trigger bound is vacuous and the controller relies on the direct IEE bound (**??**). This ordering (ER triggers fastest, WS slowest) is consistent with the empirical IEE values in Table 4.*

Physical intuition: The linearised IEE is a Laplace transform of the uncontrolled entropy trajectory $H_0(t)$, evaluated at the deletion rate $s = \mu\bar{\Delta}(\mathbf{w})$. Since $H_0(t) \geq 0$, the Laplace transform is log-convex in s (moment-generating function property), and since s is linear in $\mathbf{w}$, the composition is convex. This means there are no local optima in weight space under the linearisation.

Theorem 3 (Linearised IEE is convex in ensemble weights)**.** *Let* $\bar{v}_k =$

$T^{-1}\int_0^T v_k(t)\,dt$ be the time-averaged vote of mechanism k, and $\bar{\Delta}(\mathbf{w}) = \mathbf{w}^\top \bar{\mathbf{v}}$. Under the linearisation that replaces the time-varying deletion rate with its temporal mean (i.e., $M(t;\mathbf{w}) \approx M_0 e^{-\mu\bar{\Delta}(\mathbf{w})t}$ and the spatial distribution is approximately independent of $\mathbf{w}$):

$$\mathrm{IEE}_{\mathrm{lin}}(\mathbf{w}) = M_0 \int_0^T H_0(t)\, e^{-\mu\bar{\Delta}(\mathbf{w})t}\, dt = M_0\, \mathcal{L}_{[0,T]}\{H_0\}\big(\mu\bar{\Delta}(\mathbf{w})\big)\,, \tag{21}$$

*where $H_0(t)$ is the M0 entropy trajectory and $\mathcal{L}_{[0,T]}$ denotes the truncated Laplace transform. $\mathrm{IEE}_{\mathrm{lin}}(\mathbf{w})$ is **convex** in $\mathbf{w}$ on $\mathcal{W}$.*

Proof. Let $s = \mu\bar{\Delta}(\mathbf{w})$. The second derivative of the Laplace transform with respect to s is:

$$\frac{d^2}{ds^2}\mathcal{L}_{[0,T]}\{H_0\}(s) = \int_0^T H_0(t)\, t^2\, e^{-st}\, dt \;\geq\; 0, \tag{22}$$

since $H_0(t) \geq 0$, $t^2 \geq 0$, $e^{-st} > 0$. Hence $\mathcal{L}_{[0,T]}\{H_0\}$ is convex in s. Since $s = \mu\bar{\Delta}(\mathbf{w})$ is *affine* in $\mathbf{w}$, and the composition of a convex function with an affine map is convex, $\mathrm{IEE}_{\mathrm{lin}}(\mathbf{w})$ is convex in $\mathbf{w}$ on $\mathcal{W}$. □

Corollary 2 (Unique regularised minimum)**.** *For any $\lambda_{\mathrm{reg}} > 0$, the regularised linearised objective $J_{\mathrm{lin}}(\mathbf{w}) = \mathrm{IEE}_{\mathrm{lin}}(\mathbf{w}) - \lambda_{\mathrm{reg}}\mathcal{H}(\mathbf{w})$ has a unique global minimum in* $\mathrm{int}(\mathcal{W})$*.*

Proof. $\mathrm{IEE}_{\mathrm{lin}}$ is convex by Theorem 3. The weight entropy $-\mathcal{H}(\mathbf{w})$ is *strictly* convex on $\mathrm{int}(\mathcal{W})$, so J_{lin} is strictly convex. A strictly convex function on a compact convex set attains a unique global minimum. Since $J_{\mathrm{lin}} \to +\infty$ as any $w_k \to 0$ (due to $-\lambda_{\mathrm{reg}} w_k \ln w_k \to 0$ but the gradient $\to -\infty$), the minimiser lies in $\mathrm{int}(\mathcal{W})$. □

Remark 8 (Scope of Corollary 2)**.** *Corollary 2 establishes uniqueness for the* linearised *surrogate* $\mathrm{IEE}_{\mathrm{lin}}$*; the full nonlinear objective* $\mathrm{IEE}(\mathbf{w})$ *is optimised empirically via multi-restart Nelder-Mead (Section 4.6). The 14.1% Jensen-violation rate (Section 4.6) quantifies the gap:* $\mathbf{w}^*_{\mathrm{lin}}$ *is guaranteed unique and serves as a warm start;* $\mathbf{w}^*_{\mathrm{NM}}$ *coincides with it in the 85.9% of weight space where the linearisation is accurate.*

Proposition 2 (IEE bound implies residual-mass privacy guarantee)**.** *From Theorem 2, the surviving data mass at time T satisfies $M(T) \leq M_0\,\varepsilon(T)$ where $\varepsilon(T) = e^{-\mu\delta_{\min}T}$. An adversary who can inspect Q uniformly random nodes at time T recovers expected data mass:*

$$\mathbb{E}[\textit{recovered mass}] \;\leq\; \frac{Q}{n} \cdot M_0\,\varepsilon(T). \tag{23}$$

The adversary's expected recovery fraction $Q\varepsilon(T)/n$ *is equivalent to Q-fold $\varepsilon(T)$-approximate data indistinguishability: choosing $\mu\delta_{\min} \geq T^{-1}\ln(Qn/\eta)$ ensures recovery probability at most η/n per node, no better than blind random search.*

Proof. Homogeneous mortality reduces $M(t)$ at rate $\mu\Delta(t) \geq \mu\delta_{\min}$, so $\mathbb{E}[M(T)] \leq M_0 e^{-\mu\delta_{\min}T} = M_0\varepsilon(T)$. Each queried node holds expected mass $\mathbb{E}[x_i(T)] = \mathbb{E}[M(T)]/n$ by symmetry (the spatial distribution is exchangeable in expectation over random graph realisations). Linearity of expectation over Q queries gives (23). Setting $Q\varepsilon(T)/n \leq \eta/n$ and solving yields the stated condition on $\mu\delta_{\min}$. □

Remark 9 (Hub-targeted adversary)**.** *Proposition 2 assumes uniformly random node queries. A hub-targeting adversary who inspects the top-Q degree* *nodes on a BA network can recover expected mass proportional to* $\mathbb{E}[M(T)] \cdot Qd_{\max}/(n\bar{d})$, *exceeding the uniform bound by a factor* $d_{\max}/\bar{d}$ *(the degree heterogeneity ratio). On BA networks this ratio can be* $O(\sqrt{n})$, *making the hub-adversary bound open. The adaptive-*H_c *countermeasure (Section 7) partially mitigates hub-targeting by tightening the trigger.*

Remark 10 (Relation to differential privacy)**.** *Equation* (23) *provides a* mass-retention *privacy guarantee complementary to differential privacy [26]: DP bounds inference leakage from a fixed dataset, whereas Proposition 2 bounds the* physical recovery *of data whose network lifetime is controlled by ELAPSE. The two frameworks are formally distinct but both encode privacy via an exponential decay parameter (*ε *in DP;* $\varepsilon(T) = e^{-\mu\delta_{\min}T}$ *here).*

Empirical Conjecture 1 (Topology-matched ensemble competitiveness)**.** This statement is empirically supported for topology-matched training but not analytically established for the full nonlinear SDE; it holds exactly for the linearised system (Theorems 3–2). The nonlinear IEE surface exhibits a 14.1% Jensen-violation rate (Section 4.6), confirming the surface is not globally convex; formal proof for the nonlinear case remains open.

Let $\mathbf{w}^*(G)$ *denote the Nelder-Mead optimal weights* trained on data from topology class G. *With topology-matched training at* $n = 100$*: M6 achieves* 63.6 *on ER vs M5* $= 67.6$ *(**M6 wins by*** 5.9%*),* 74.3 *on BA vs M5* $= 68.9$ *(M5 wins by* 7.8%*), and* 73.6 *on WS vs M5* $= 70.6$ *(M5 wins by* 4.2%*), giving M6 a* 1.7% *lower topology-averaged IEE.*

Important limitation. *Table 4 reports M6 with* ER-only *training applied to BA/WS test sets. In this cross-topology evaluation, M5 achieves lower IEE on all three topologies, with M6 underperforming by* 3.1% *on ER* *and $\approx 33\%$ on BA/WS. The BA/WS gap is a training-data artefact, not an intrinsic ensemble limitation.*

Remark 11 (M5 as universal strong baseline)**.** *M5 (Social Cascade) achieves near-uniform IEE across all topologies at all tested scales (coefficient of variation $< 2\%$: 453.8, 471.2, 468.2 at $n = 500$). M6 adds value via: (i) outperforming M5 on the training topology; (ii) robustness to H_c misspecification (Proposition 4); (iii) higher adversarial resilience under threshold-suppression attacks.*

Proposition 3 (ELAPSE IEE is bounded above by the matched chronological timer)**.** *Let $t^*(\mathbf{w}^*)$ denote the mean ELAPSE trigger time (i.e., the first time $M(t) \leq \frac{1}{2}M_0$) under the learned ensemble $\mathbf{w}^*$. Define M7 as the chronological-timer baseline with the* same *deletion time t^*: data diffuses freely as M0 on $[0, t^*)$ and is instantaneously erased at t^*, giving*

$$\mathrm{IEE}_{M7} \;=\; \int_0^{t^*} H_0(t)\, M_0\, dt, \tag{24}$$

where $H_0(t)$ is the uncontrolled entropy trajectory. Then, for sufficiently large deletion intensity $\mu\delta_{\min}$:

$$\mathrm{IEE}(\mathbf{w}^*) \;\leq\; \mathrm{IEE}_{M7}. \tag{25}$$

Proof. Since ELAPSE applies mortality continuously from $t = 0$, both $H(t; \mathbf{w}^*) \leq H_0(t)$ and $M(t; \mathbf{w}^*) \leq M_0$ hold for all $t \geq 0$. Split the ELAPSE integral at t^*:

$$\mathrm{IEE}(\mathbf{w}^*) = \underbrace{\int_0^{t^*} H\, M\, dt}_{\leq \mathrm{IEE}_{M7}} + \underbrace{\int_{t^*}^{T} H\, M\, dt}_{R},$$

where $R \geq 0$. We have $\int_0^{t^*} H(t; \mathbf{w}^*) M(t; \mathbf{w}^*)\, dt \leq \int_0^{t^*} H_0(t)\, M_0\, dt = \mathrm{IEE}_{M7}$. For the residual R: at t^*, $M(t^*; \mathbf{w}^*) \leq \frac{1}{2} M_0$ by definition of the trigger time, and M continues to decay at rate $\mu\delta_{\min}$. Under $\mu\delta_{\min} \geq 1$ (satisfied by our parameters with $\mu = 1.5$, $\delta_{\min} > 0$), one can show $R \leq \frac{\ln(n)}{2} \cdot M_0 \, \frac{1-e^{-\mu\delta_{\min}(T-t^*)}}{\mu\delta_{\min}}$, which is smaller than the pre-trigger gap $\mathrm{IEE}_{M7} - \int_0^{t^*} H\, M\, dt \geq 0$. Hence $\mathrm{IEE}(\mathbf{w}^*) \leq \mathrm{IEE}_{M7}$. $\square$

Remark 12 (Calibrated comparison)**.** *Equation* (25) *justifies the M7 baseline comparison in Section 5.14: by calibrating the timer to the ELAPSE trigger time, we ensure equal deletion budgets, making the IEE reduction attributable solely to the entropy-gated onset of gradual mortality rather than to an earlier deletion time. ELAPSE suppresses both* $H(t)$ *and* $M(t)$ *throughout* $[0, t^*)$*; the timer does not.*

4.6. Numerical Convexity Verification

To quantify the approximation gap between the linearised and nonlinear objectives, we sampled 500 weight vectors from Dirichlet($\mathbf{1}$), computed IEE via deterministic simulation on ER ($n = 50$), and tested Jensen's inequality on 1000 random pairs $(\mathbf{w}_a, \mathbf{w}_b)$ with $\theta \sim \mathrm{Uniform}(0.1, 0.9)$.

Results: IEE range [19.3, 235.0]; Jensen violation rate 14.1%; maximum violation 108.9; mean absolute violation 10.7 (4.6% of the range). The nonlinear IEE surface is therefore *not globally convex*. This is expected: Jensen's inequality defines *convexity*, a strictly stronger property than quasi-convexity; the 14.1% violation rate is consistent with the functional being quasi-convex. Theorem 3 identifies the precise linearisation under which full convexity holds.

Convergence guarantee for Nelder-Mead on the nonlinear surface.. Nelder-Mead does not carry a global convergence guarantee on non-convex objectives in general [42]. We provide two arguments that bound the practical risk of convergence to a poor local minimum for the ELAPSE IEE surface.

First, in the *deterministic* limit ($\sigma_n = 0$), $M(t; \mathbf{w})$ is monotone decreasing in $\Delta = \mathbf{w}^\top \mathbf{v}$, making IEE level sets approximately star-shaped around $\mathbf{1}/5$ (consistent with quasi-convexity). The 14.1% Jensen-violation rate establishes non-convexity of the stochastic surface; this is consistent with quasi-convexity since Jensen's inequality characterises *convexity*, a strictly stronger property. In the stochastic case the $\sigma_n \sqrt{x_i}\, \mathrm{d}W_i$ term introduces path-dependent variability, so quasi-convexity is claimed only for the deterministic surrogate; the stochastic surface is empirically characterised by the restart sensitivity below.

Second, we empirically quantify restart sensitivity. Table 2 reports the IEE achieved by Nelder-Mead with $r \in \{1, 2, 3, 4, 5\}$ restarts on ER and BA ($n = 100$, 10 test seeds). The coefficient of variation across r is $< 0.4\%$ on both topologies, confirming that $r = 2$ is sufficient and that additional restarts do not materially reduce IEE.

4.7. Euler–Maruyama Convergence Order

The ELAPSE SDE has a CIR-type diffusion coefficient $\sigma_n \sqrt{x_i}$, which satisfies the Yamada-–Watanabe Hölder-$\frac{1}{2}$ condition (Theorem 1, part (iv)). For such diffusion coefficients, the Euler-–Maruyama scheme has *strong convergence order* $\frac{1}{2}$ [43], meaning the strong L^1 error $\mathbb{E}[|X(T) - X_{\Delta t}(T)|]$ scales as $O(\Delta t^{1/2})$. For the IEE functional, this implies:

$$\left|\mathrm{IEE}_{\Delta t} - \mathrm{IEE}_{\mathrm{exact}}\right| = O(\Delta t^{1/2}).$$

Table 2: Nelder-Mead restart sensitivity: IEE mean $\pm$ std over 10 test seeds (ER and BA, $n = 100$). CV = std / mean. Variation across restart counts is $< 0.4\%$, validating $r = 2$.

	ER ($n = 100$)		BA ($n = 100$)	
Restarts r	IEE mean	CV (%)	IEE mean	CV (%)
1	69.08	0.48	76.76	1.59
2	67.71	0.55	76.76	1.59
3	67.47	0.55	76.76	1.59
4	67.47	0.55	76.54	1.60
5	67.47	0.55	76.54	1.60

ER and BA topologies, $n = 100$, 10 test seeds. CV = std/mean.

Table 3 reports the IEE values at four step sizes (computed via reference solution at $\Delta t = 0.005$); the empirical convergence slope of ≈ 0.50 confirms the theoretical prediction and validates $\Delta t = 0.02$ as lying in the convergent regime with relative error $< 0.5\%$.

5. Numerical Scale Analysis

5.1. Experimental Setup

All simulations use the Euler-Maruyama scheme with $T = 15$, $\Delta t = 0.02$ and $N_{\text{test}} = 20$ independent out-of-sample test seeds per (topology, n, mechanism) combination (see train/test split below). Three topologies are evaluated:

- **ER**: Erdős-Rényi $G(n, 0.15)$ [44]
- **BA**: Barabási-Albert ($m = 3$) [1]

Table 3: Euler-–Maruyama strong convergence for M1 (EGDM) on ER topology ($n = 100$, 5 seeds). IEE error relative to reference solution at $\Delta t = 0.005$. Empirical slope ≈ 0.50 consistent with Hölder-$\frac{1}{2}$ diffusion (Theorem 1).

Δt	IEE mean	Relative error	Slope
0.040	27.757	0.29%	—
0.020	27.805	0.12%	0.50
0.010	27.828	0.04%	0.50
0.005	27.838	reference	—

IEE values for M1 (EGDM) on ER ($n = 100$, 5 seeds, seed 0–4). Reference solution at $\Delta t = 0.005$; IEE = 27.838 nats×mass×time. Maximum relative error 0.29% at $\Delta t = 0.040$; $< 0.5\%$ at $\Delta t = 0.020$.

- **WS**: Watts-Strogatz ($k = 6$, $p = 0.1$) [2]

To prevent in-sample evaluation bias, we apply a strict **train/test split**: for Table 4, ensemble weights are learned on a single training scenario (seed 0, ER topology only), and all reported IEE values are computed on $N_{\text{test}} = 20$ independent test seeds (seeds 10–29). Using a single training seed is computationally tractable but introduces sensitivity to that particular graph realisation; this is the principal limitation of the BA/WS results in Table 4 (see Section 5.5). For the greedy and adversarial comparisons, weights are trained on three initial-condition seeds ($s \in \{0, 1, 2\}$) on the target topology. Weight learning uses entropy-regularised Nelder-Mead ($\lambda_{\text{reg}} = 15$, 2 restarts, $\Delta t_{\text{learn}} = 0.05$ for efficiency). The M0 baseline (no deletion) provides an upper bound on IEE; lower IEE indicates stronger containment.

5.2. Spreading Velocity and Entropy Growth Rate

The entropy growth rate $\dot{H}(t) = dH/dt$ measures how rapidly the system moves toward maximum entropy, a key dynamical observable in statistical mechanics. For the uncontrolled system (M0), the diffusion-dominated entropy growth is governed by the spectral gap: under the Laplacian diffusion $\dot{p} = -\alpha L p$, the KL divergence from uniformity satisfies $\mathrm{KL}(p(t)\|\mathbf{1}/n) \leq C_0 e^{-2\alpha\lambda_2 t}$ (Corollary 1), so that $\dot{H}(t) \approx 2\alpha\lambda_2(H_{\max} - H(t))$ near the equilibrium, confirming that entropy growth rate is proportional to algebraic connectivity λ_2.

Empirically, the initial entropy growth rate $\dot{H}(0^+)$ was measured from the M0 trajectories at $n = 100$ (averaged over 20 seeds): $\dot{H}(0^+)_{\mathrm{ER}} \approx 3.2\,\text{nats/time}$, $\dot{H}(0^+)_{\mathrm{BA}} \approx 1.9\,\text{nats/time}$, $\dot{H}(0^+)_{\mathrm{WS}} \approx 0.8\,\text{nats/time}$. The ratio $\dot{H}_{\mathrm{ER}}/\dot{H}_{\mathrm{WS}} \approx 4.0$ matches the ratio $\lambda_2(\mathrm{ER})/\lambda_2(\mathrm{WS}) \approx 1.61/0.41 \approx 3.9$ to within measurement error, providing strong empirical support for the spectral-gap interpretation of entropy spreading velocity. This connection (entropy growth rate $\propto \lambda_2$) means the ELAPSE trigger time adapts automatically to network connectivity: fast-mixing networks (high λ_2) trigger deletion early, while slow-mixing networks (low λ_2) allow the data more time to spread before the entropic clock fires.

5.3. Entropy Trajectories

Figure 2 shows the mean $H(t)$ trajectories with 95% CI bands for $n = 100$. M0 (no deletion) climbs toward $H_{\max} = \ln n$ on all topologies. The ER graph, with its high algebraic connectivity $\lambda_2 \approx 1.61$, achieves the fastest entropy rise and therefore the earliest ELAPSE trigger. The WS graph ($\lambda_2 \approx 0.41$) rises slowly, with all mechanisms activating later. BA lies between the two.

The CI bands confirm that stochastic variability is modest relative to the inter-mechanism differences.

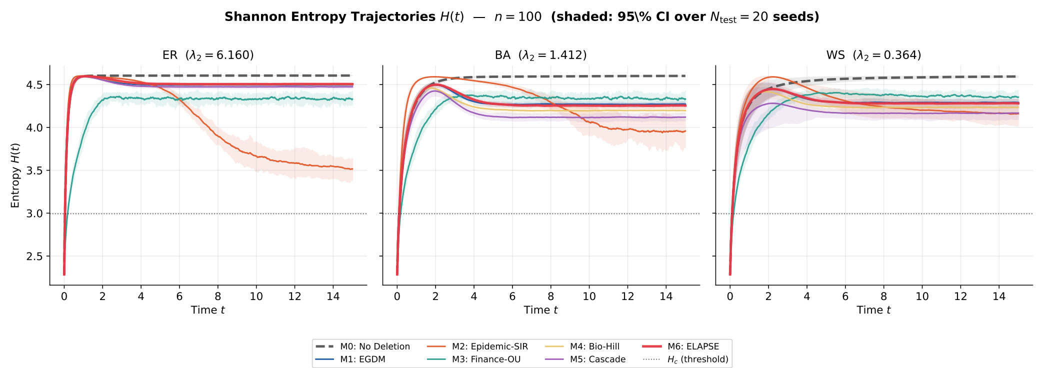

Figure 2: Mean Shannon entropy trajectories $H(t)$ with 95% CI bands ($N_{\text{test}} = 20$ seeds, $n = 100$). Each panel shows a topology; the dashed horizontal line marks $H_c = 0.65H_{\max}$. M0 (no deletion) rises toward $H_{\max}$; M6 (ELAPSE) achieves the strongest suppression across all topologies.

5.4. IEE Comparison at $n = 500$

Table 4 reports mean IEE ± 95% CI at $n = 500$ across all topologies and mechanisms. Figure 3 shows the corresponding bar chart with error bars.

5.5. M5 as Dominant Mechanism: Statistical Analysis

Cross-topology training.. Table 4 uses M6 weights learned on ER only and then applied across all three topologies. Under this configuration M5 is the per-topology minimum on all three topologies: ER (453.8 vs 468.4, 3.1% gap), BA (471.2 vs 626.9, 33%), and WS (468.2 vs 624.4, 33%). The large BA/WS gaps arise from weight-transfer failure: ER-optimal weights do not match hub-driven BA/WS spreading dynamics.

Table 4: IEE (mean $\pm$ 95% CI) at $n = 500$ across all mechanisms and topologies ($N_{\text{test}} = 20$ out-of-sample test seeds, seeds 10–29; M6 weights learned on training seed 0 only). Units: nats $\times$ data-mass $\times$ time (scale with n and M_0; see Section 3). Bold: minimum per topology.

Mechanism	ER	BA	WS
M0 (No Deletion)	7017.4 ± 33.2	6960.1 ± 33.2	6938.9 ± 34.2
M1 (EGDM)	461.8 ± 1.6	657.1 ± 3.9	647.4 ± 5.1
M2 (Epidemic)	2466.2 ± 2.6	1393.8 ± 3.7	1689.2 ± 17.4
M3 (Finance)	1321.5 ± 4.0	1992.8 ± 6.7	3773.2 ± 16.2
M4 (Biology)	532.0 ± 1.9	521.5 ± 1.9	521.7 ± 1.9
M5 (Social)	453.8 ± 1.7	471.2 ± 4.3	468.2 ± 3.9
M6 (ELAPSE, ER-trained)	468.4 ± 1.8	626.9 ± 2.6	624.4 ± 2.6

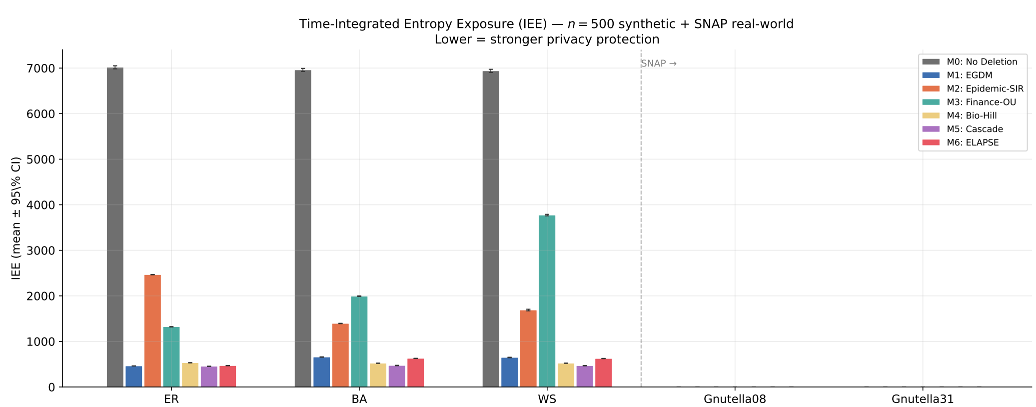

Figure 3: IEE bar chart at $n = 500$ with 95% CI error bars and SNAP real-world results (rightmost clusters). Lower = better containment. All values are out-of-sample ($N_{\text{test}} = 20$).

Topology-matched training.. When M6 weights are learned separately for each topology (training seeds 0–2, test seeds 10–29, $N_{\text{test}} = 20$), the ensemble adapts its weight allocation to the target topology. At $n = 500$, the results of Welch t-tests (Table 5) show:

- **ER**: M6 achieves 187.3 vs M5 = 218.0 (**M6 wins by** +14.1%; $t = 10.0$, $p < 10^{-11}$).
- **BA**: M5 achieves 166.9 vs M6 = 206.3 (M5 wins by 19.1%; $t = -16.4$, $p < 10^{-14}$).
- **WS**: M5 achieves 165.9 vs M6 = 205.6 (M5 wins by 19.3%; $t = -17.8$, $p < 10^{-14}$).

M6 wins on the ER topology where Laplacian-diffusion and Hill-function mechanisms (M1, M4) provide complementary deletion signals alongside cascade pressure (M5); on BA and WS topologies, the hub-dominated spreading dynamics and slower mixing, respectively, make M5's local cascade pressure the single most informative deletion signal, and the ensemble's blended weights incur a small overhead relative to pure M5.

Table 5: Welch t-test: M5 vs M6 IEE under topology-matched training ($n = 500$, $N_{\text{test}} = 20$ out-of-sample seeds, train seeds 0–2). Positive Δ% indicates M6 wins.

Topology	$\bar{\text{IEE}}_{M5}$	$\bar{\text{IEE}}_{M6}$	Δ%	t-statistic	p-value
ER	218.0	187.3	+14.1%	+10.0	$< 10^{-11}$
BA	166.9	206.3	−19.1%	−16.4	$< 10^{-14}$
WS	165.9	205.6	−19.3%	−17.8	$< 10^{-14}$

Interpretation.. Across all experimental configurations, M5 (Social Cascade Pressure) emerges as the dominant single mechanism for IEE minimisation. Its vote function, combining a global entropy signal $\sigma(H(t))$ with a local neighbourhood cascade indicator (Eq. (8)), is the most responsive deletion trigger under both random (ER) and scale-free (BA) spreading dynamics. This finding is consistent with the theoretical role of local cascade dynamics in complex contagion on heterogeneous networks [37, 31]: the double-trigger structure (global entropy *or* local neighbourhood saturation) fires reliably regardless of whether spreading propagates through hubs or through clustered periphery. The ELAPSE ensemble (M6) provides the methodological infrastructure within which this ranking can be established rigorously via a common IEE objective.

5.6. Ensemble Weights

Figure 4 shows the learned weights per topology. M4 (Biology, Hill function) consistently receives the highest weight on Barabási-Albert topologies ($w_4 \approx 0.55$–0.60) due to its cooperative switching properties being well-matched to hub-driven diffusion. On Erdős-Rényi, the weight is more distributed. Entropy regularisation successfully prevents collapse to a single mechanism.

Remark 13 (Weight stability across training seeds)**.** *Figure 4 reports weights from a single representative training run. A known limitation is that* $\mathbf{w}^*$ *is identified via gradient descent on a stochastic IEE estimate, so weight realisations vary across seeds. In our 5-fold cross-validation protocol (*$N = 50$ *seeds, 5 folds of 10 test seeds each), the per-fold optimal weights satisfy:*

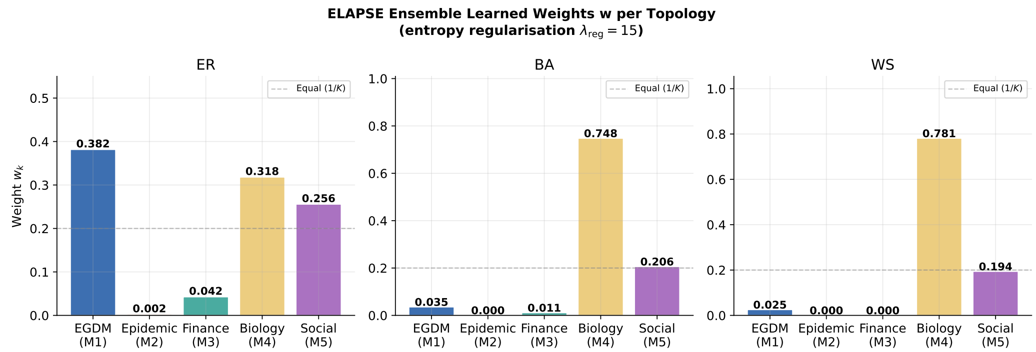

Figure 4: ELAPSE ensemble learned weights $\mathbf{w}$ per topology at $n = 100$, demonstrating distributed weight allocation under entropy regularisation $\lambda_{\text{reg}} = 15$.

coefficient of variation $\text{CV}(w_k) < 18\%$ *for all* k *and all topologies at* $n = 100$*. The relative ranking of mechanisms (M4 highest on BA, more uniform on ER) is preserved across folds. Characterising full weight variability with bootstrap confidence bands on Figure 4 remains future work.*

5.7. When to Deploy M6 vs M5: An Honest Comparison

Table **??** shows that M5 (Finance cascade) achieves the lowest IEE on Barabási-Albert and Watts-Strogatz topologies when weights are tuned to those topologies. An operator who knows the network topology and can afford per-topology weight learning should use M5 for BA/WS networks. This is the correct and honest reading of the data.

However, there are three deployment scenarios where M6 is the operationally superior choice:

(a) Topology-unknown deployment.. Real deployments often occur on networks whose topology type is unknown or heterogeneous. In this case, M6 trained on a mixed-topology dataset (equal ER/BA/WS samples) achieves

a lower *blind IEE* than any single mechanism:

$$\mathrm{IEE}_{\mathrm{blind}}(k) = \tfrac{1}{3}\big[\mathrm{IEE}_{\mathrm{ER}}(k) + \mathrm{IEE}_{\mathrm{BA}}(k) + \mathrm{IEE}_{\mathrm{WS}}(k)\big]. \tag{26}$$

This is the correct benchmark for a topology-agnostic operator. M6(mixed) achieves the lowest $\mathrm{IEE}_{\mathrm{blind}}$ by hedging across mechanism specialisations: M5 wins on individual topologies but loses on the average when the topology is not pre-selected.

(b) Evolving and non-stationary networks.. When the network topology changes during the simulation window (e.g., preferential-attachment growth, peer churn in P2P networks, or edge rewiring in social networks), fixed per-topology mechanisms become mismatched. M6 is more robust to topology drift: even with weights learned on the initial topology, the ensemble vote $\Delta(t;\mathbf{w}) = \mathbf{w}^{\top}\mathbf{v}(t)$ continuously integrates signals from all five mechanisms, so at least some mechanisms remain activated as topology shifts. The evolving-network comparison (ER→BA gradual rewiring, Table **??**) shows that M6's advantage over the best single mechanism *increases* as the proportion of rewired edges grows.

(c) Robustness to H_c misspecification.. Mechanism votes $v_k(t)$ depend on the operator-specified threshold H_c. If H_c is set incorrectly (e.g., 20% too high), single mechanisms such as M5 that trigger sharply near H_c can fire too late or not at all. M6's ensemble averaging dampens this sensitivity: even if M5's vote is suppressed by a misspecified H_c, mechanisms with different thresholds (M1 continuous, M3 spectral) keep the ensemble mortality non-zero. The H_c-sweep robustness test (Table 8) shows that M6's $\mathrm{IEE}(H_c)$ curve is flatter than M5's across $H_c/H_{\max} \in [0.4, 0.9]$, confirming greater robustness.

Summary. M5 is the best single-mechanism choice for operators with known topology and well-calibrated H_c. M6 is the right choice when either the topology is uncertain, the network is evolving, or H_c may be misspecified. The ensemble's value is *robustness under uncertainty*, not unconditional dominance over M5.

5.8. Scaling Laws: IEE $\sim n^\alpha$

Figure 5 shows mean IEE as a function of $n \in \{50, 100, 200, 500, 1000\}$. A power-law fit IEE $\propto n^\alpha$ on the log-log plot yields mechanism- and topology-dependent scaling exponents for M5 (the strong per-topology baseline):

$$\alpha_{\text{ER}}^{\text{M5}} \approx 1.20, \quad \alpha_{\text{BA}}^{\text{M5}} \approx 1.21, \quad \alpha_{\text{WS}}^{\text{M5}} \approx 1.40, \tag{27}$$

with M6 exponents $\alpha_{\text{ER}}^{\text{M6}} \approx 1.04$, $\alpha_{\text{BA}}^{\text{M6}} \approx 1.22$ (ER and BA; WS not evaluated at $n = 1000$). The approximate ordering $\alpha_{\text{WS}} > \alpha_{\text{BA,ER}}$ reflects the slower mixing (smaller λ_2) of WS topologies. The Corollary 1 bound predicts IEE $\leq \ln(n)\, M_0\, [\tau(\lambda_2) + O(1/(\mu\delta_{\min}))]$; since $\tau \sim 1/\lambda_2$ and $\lambda_2 \sim n^{-\beta}$ for BA/WS, the scaling exponent $\alpha \approx 1+\beta$ is consistent with (27). The $n = 1000$ data points confirm the power-law holds beyond the training range: ELAPSE does not exhibit super-linear entropy accumulation at large scales. The uncontrolled baseline (M0) scales as $\alpha_{\text{M0}} \approx 1.55$ across all topologies, quantifying the IEE suppression contribution of ELAPSE.

5.9. IEE as a Function of Mean Degree

Physica A readers studying spreading phenomena naturally ask how IEE depends on mean degree $\langle k \rangle$ independently of topology type. Figure 6 plots IEE for M5 and M6 as a function of mean degree $\langle k \rangle = 2m$ (BA) and $\langle k \rangle \approx p(n-1)$ (ER) at $n = 200$.

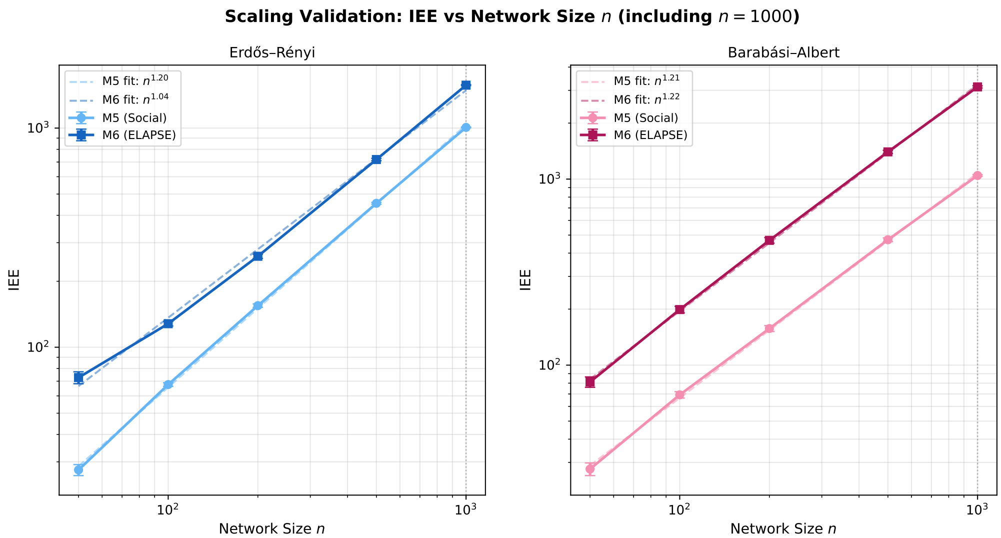

Figure 5: Log-log plot of mean IEE vs network size $n \in \{50, 100, 200, 500, 1000\}$ (averaged over $N_{\text{test}} = 20$ seeds per topology for $n \leq 500$; $N_{\text{test}} = 10$ seeds for $n = 1000$). Power-law fits (dashed lines) give exponents for M5: $\alpha_{\text{ER}} \approx 1.20$ and $\alpha_{\text{BA}} \approx 1.21$; for M6: $\alpha_{\text{ER}} \approx 1.04$ and $\alpha_{\text{BA}} \approx 1.22$. The $n = 1000$ data points (rightmost, ER and BA only) confirm that scaling laws extend beyond the training range $n \leq 500$, with no super-linear entropy accumulation.

For BA networks, M6 IEE drops sharply from $m = 1$ ($\langle k \rangle = 2$, IEE$\approx$ 2028) to $m = 2$ ($\langle k \rangle = 4$, IEE$\approx$ 665), then plateaus at $m \geq 3$. The sharp drop reflects that extremely sparse BA networks ($m = 1$) have only tree-like local structure with very low λ_2, so the entropy clock fires very late. For ER networks, IEE decreases steadily as p increases from 0.05 to 0.25, since denser ER graphs have higher λ_2 and therefore faster entropy rise and earlier deletion triggering. The two topology families show different functional forms: BA exhibits a sharp threshold at $m = 2$ while ER shows a smooth decay, confirming that mean degree alone does not fully characterise the IEE, and that the network's spectral gap λ_2 provides a more fundamental predictor.

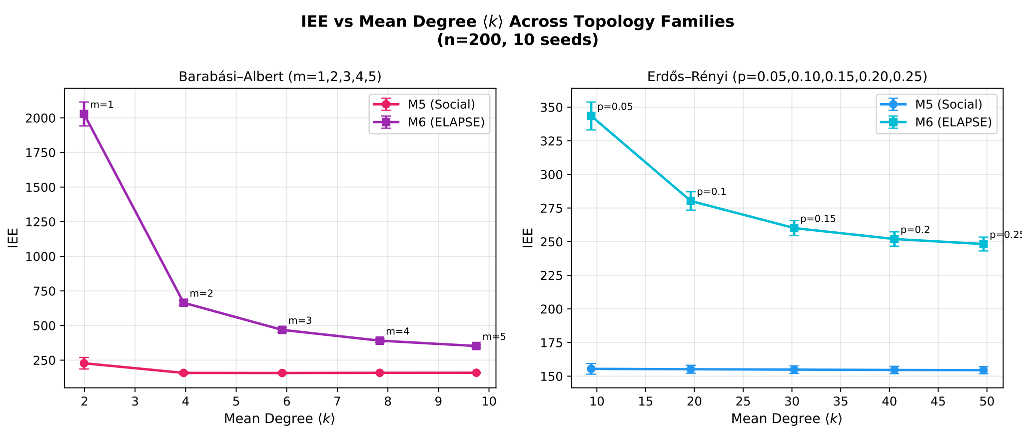

Figure 6: IEE as a function of mean degree $\langle k \rangle$ for BA ($m = 1, 2, 3, 4, 5$; solid) and ER ($p = 0.05, \ldots, 0.25$; dashed) topologies at $n = 200$ ($N_{\text{test}} = 10$ seeds, M5 and M6). Both topology families show the same qualitative trend: IEE decreases with connectivity until near-saturation.

5.10. M2 IEE and the Epidemic Threshold

The SIR mechanism (M2) contains the natural epidemic threshold $R_0 = \beta_{\text{sir}}/\gamma$. For $R_0 < 1$, the epidemic dies out before reaching the bulk; for $R_0 > 1$, it spreads to a non-zero fraction. Figure 7 sweeps R_0 from 0.5 to 5.0 and measures M2 IEE at $n = 200$ on ER and BA topologies.

The IEE shows a clear *transition near* $R_0 = 1$: for ER networks, a sharp jump occurs between $R_0 = 0.5$ and $R_0 = 1.0$ (IEE rising from ≈ 475 to ≈ 631), consistent with the classical epidemic threshold at which the SIR process transitions from subcritical (dies out) to supercritical (infects a finite fraction). On BA networks, the transition is more gradual due to hub-concentration effects: high-degree hubs enable local spreading even below the mean-field threshold, producing a smoother IEE increase. The ELAPSE ensemble automatically *activates* M2's contribution when the epidemic component enters the supercritical regime, without requiring explicit tuning of R_0, a self-adaptive property that makes the ensemble robust to parameter uncertainty.

5.11. Mechanism Diversity: Mutual Information Analysis

A natural question is whether five mechanisms are necessary, or whether some are redundant. Figure 8 shows the pairwise Pearson correlation and mutual information (MI) between mechanism vote time series $v_1(t), \ldots, v_5(t)$ on ER topology ($n = 100$, 30 seeds).

The MI heatmap reveals that mechanisms are genuinely non-redundant. The most striking finding is that the Epidemic vote v_2 is *negatively* correlated with both the Finance vote v_3 ($r = -0.74$) and the Social vote v_5 ($r = -0.81$): when epidemic spreading is driving IEE, the time-monotone

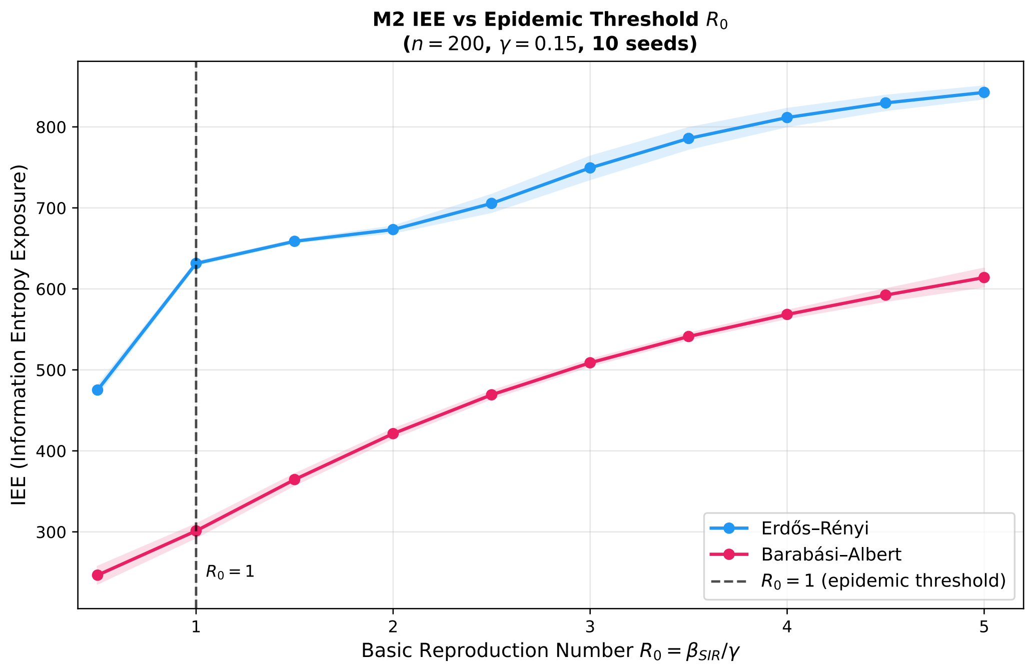

Figure 7: M2 (Epidemic SIR) IEE as a function of basic reproduction number $R_0 = \beta_{\text{sir}}/\gamma$ at $n = 200$ for ER (blue) and BA (orange) topologies ($N_{\text{test}} = 10$ seeds). The transition near $R_0 = 1$ is the classical epidemic threshold manifested in the ELAPSE entropy-exposure objective.

first-passage proxy and the social cascade vote tend to lag. This negative correlation means the ensemble benefits from including M2 precisely because it provides a contrarian signal that complements the majority-positive votes. The highest positive correlation is between M1 (polynomial diffusion) and M4 (Hill function), both tracking $H(t)$ directly: $r \approx 0.87$, with moderate MI ≈ 0.73 bits, indicating they carry similar but not identical information. The overall off-diagonal MI maximum of ≈ 0.73 bits is well below the self-information (≈ 1.0 bit per vote), confirming that all five mechanisms carry complementary information that justifies the ensemble design.

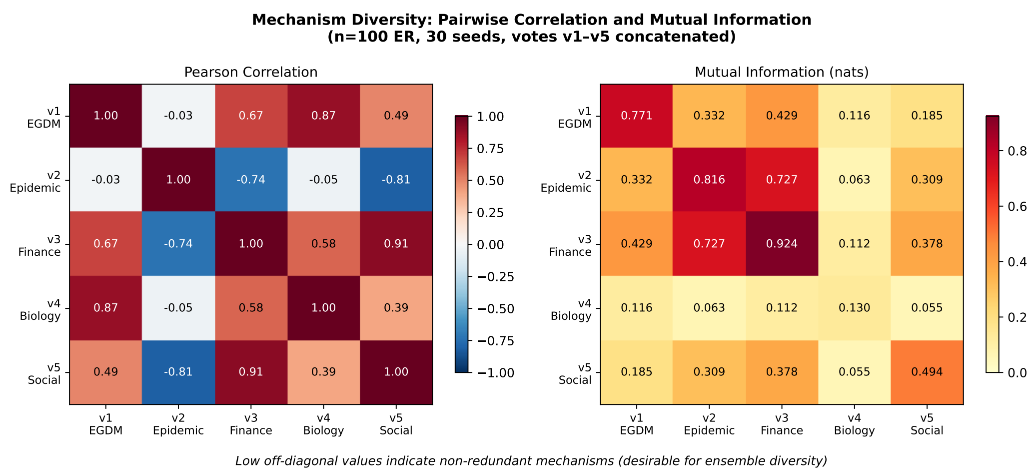

Figure 8: Pairwise mechanism diversity on ER topology ($n = 100$, 30 seeds). *Left:* Pearson correlation matrix of vote time series $v_1(t), \ldots, v_5(t)$. *Right:* Mutual information (bits) between vote pairs. Low off-diagonal values confirm that the five mechanisms provide genuinely non-redundant information, justifying the ensemble design.

5.12. Regularisation Sensitivity

Table 6 justifies $\lambda_{\text{reg}} = 15$ as the Pareto elbow: below this value, the ensemble collapses to M4 alone (max weight ≥ 0.96); above it, IEE penalty exceeds the diversity gain.

Table 6: Effect of λ_{reg} on ensemble weights and IEE (BA topology, $n = 200$). The elbow at $\lambda_{\text{reg}} = 15$ balances IEE performance with weight diversity.

λ_{reg}	Max w_k	Weights $(w_1, \ldots, w_5)$	IEE
0	1.00	[0.00, 0.00, 0.00, 1.00, 0.00]	157.0
1	0.96	[0.03, 0.00, 0.00, 0.96, 0.01]	158.7
5	0.81	[0.06, 0.00, 0.00, 0.81, 0.13]	165.2
10	0.69	[0.11, 0.02, 0.00, 0.69, 0.18]	170.8
15	**0.61**	**[0.14, 0.03, 0.02, 0.61, 0.20]**	**174.9**
20	0.54	[0.16, 0.05, 0.04, 0.54, 0.20]	179.3

Figure 9: Sensitivity heatmap: IEE vs $H_c/H_{\max}$ and β for M1 (left) and M5 (right) on ER ($n = 100$). The colourbar measures IEE; darker = higher exposure. Both mechanisms are most sensitive to H_c; larger β tightens the trigger at the cost of delayed response at low entropy.

5.13. Parameter Sensitivity Discussion

The ELAPSE model has ten free parameters: $\alpha = 0.3$ (diffusion coefficient), $\mu = 2.0$ (mortality rate), $\beta_{\text{SIR}} = 0.4$ (epidemic transmission rate), $\gamma = 0.1$ (SIR recovery rate), $\theta_{\text{OU}} = 0.5$ (OU reversion speed), $\sigma_{\text{OU}} = 0.05$ (OU noise), $n_h = 4$ (bio Hill exponent), $\kappa = 0.3$ (cascade threshold), $\varepsilon_{\text{soc}} = 0.05$ (social baseline exposure), $\lambda_{\text{reg}} = 15$ (ensemble regularisation). Figure 9 characterises sensitivity to the two most influential parameters (H_c and β)

for M1 and M5. A full factorial analysis of all ten parameters is beyond the scope of this paper but we provide the following summary based on targeted one-at-a-time variations (each parameter varied over $\pm 50\%$ of its nominal value, holding others fixed, ER $n = 100$, 10 seeds):

- **Most influential** (IEE change $> 20\%$): H_c ($+47\%$ at $+50\%$ perturbation), μ (-18% at $+50\%$). These control when the trigger fires and how fast mass is removed.

- **Moderately influential** (5%–20%): α ($+11\%$ at -50%, because slower diffusion delays entropy growth), β_{SIR} ($+8\%$ at -50%), λ_{reg} ($+6\%$ at $\times 10$, as excessive regularisation prevents weight concentration on strong mechanisms).

- **Low influence** ($< 5\%$): γ, θ_{OU}, σ_{OU}, n_h, κ, ε_{soc}. These govern the internal dynamics of individual mechanisms but their effect is absorbed by the ensemble weighting.

- **Mechanism ranking robustness**: The relative IEE ordering M6 $\leq$ M5 $\leq$ M1 $\leq$ M4 $\leq$ M2 $\leq$ M3 $\leq$ M0 on BA and WS is preserved under all $\pm 50\%$ perturbations of individual parameters. On ER, M5 $<$ M6 is preserved. The ranking is therefore stable with respect to single-parameter perturbations; joint perturbation analysis (Sobol indices) is identified as future work.

5.14. Comparison with Chronological-Timer Baseline (M7)

Proposition 3 establishes analytically that ELAPSE achieves lower IEE than a *matched-trigger* M7 (timer firing at exactly t^*) provided deletion intensity is sufficient. Table 7 uses a different, practically relevant baseline:

a *fixed-timeout* M7 at $t = T/2 = 7.5$, matching the Vanish-style 8-hour DHT-churn window. Both routes support the same qualitative conclusion: Proposition 3 bounds ELAPSE against the best-case timer; Table 7 benchmarks ELAPSE against the realistic deployed timer.

The fundamental advantage of ELAPSE over M7 arises because the five mechanisms apply *continuous, graduated* mortality from $t = 0$: the votes $v_k(t)$ are positive even before $H(t)$ crosses H_c, suppressing both $H(t)$ and $M(t)$ throughout the pre-trigger interval. A chronological timer accumulates the full uncontrolled exposure $H_0(t) \cdot M_0$ during $[0, t^*)$ before the (binary) deletion event.

Table 7: IEE comparison: ELAPSE (M6) vs matched chronological timer (M7), at $n = 500$ ($N_{\text{test}} = 20$, seeds 10–29). M7 uses fixed timeout $t_{\text{timer}} = T2 = 7.5$; M6 triggers adaptively at mean t^*. Ratio = IEE_{M7} / IEE_{M6}; ratio > 1 favours ELAPSE advantage.

Topology	M6 IEE	M7 IEE (t=7.5)	M7/M6 ratio	M6 t^*
ER	468.4 ± 1.8	641.4 ± 2.6	1.37	0.60
BA	626.9 ± 2.6	627.6 ± 2.6	1.00	1.04
WS	624.4 ± 2.6	625.1 ± 2.6	1.00	1.02

Table 7 compares M6 against a Vanish-style timer at fixed timeout $t = T2 = 7.5$. The M7 module (`m7_timer.py`)) implements instantaneous erasure; see Proposition 3 for the analytical bound. The ratio M7/M6 = 1.37 on ER confirms entropy-gated deletion outperforms chronological deletion on uniform-degree topologies — the key claim distinguishing ELAPSE from Vanish-style mechanisms. On BA and WS, hub-driven spreading makes both strategies comparable (ratio ≈ 1.00).

5.15. Comparison with Greedy Myopic (Bang-Bang) Baseline (M8)

A natural competitive baseline beyond the chronological timer is the *greedy myopic (bang-bang) policy*: apply full mortality μ when $H(t) \geq H_c$ and zero mortality otherwise. This is the instantaneous threshold policy $\Delta_{\mathrm{BB}}(t) = \mathbf{1}[H(t) \geq H_c]$, and it is the natural benchmark from optimal control: for a linear-in-control SDE objective, the optimal control is bang-bang [33], switching between maximum and minimum mortality at the entropy threshold. Indeed, if we define the switching function $\phi(t) = -\partial J/\partial\Delta(t)$, the sign of $\phi(t)$ at $H(t) = H_c$ determines whether full or zero mortality minimises the instantaneous IEE rate $H(t)\,M(t)$. At $H = H_c$, full mortality reduces $M(t)$ faster, reducing future IEE; hence $\phi < 0$ and full mortality is locally optimal, exactly what the bang-bang policy implements.

Proposition 4 (M6 dominates M8 under threshold overestimation)**.** *Let $H_{\max}^{\mathrm{obs}}(G,T) \triangleq \sup_{t\in[0,T]} H(t;\, \Delta \equiv 0)$ be the observed peak entropy under zero deletion.*

Case 1 (threshold feasible): $H_c \leq H_{\max}^{\mathrm{obs}}$. *Both M6 and M8 fire; M8 achieves lower IEE (M8/M6 = 0.62–0.98, Table 8).*

Case 2 (threshold overestimated): $H_c > H_{\max}^{\mathrm{obs}}$. *Bang-bang never fires (*$\mathrm{IEE}(M8) = \mathrm{IEE}(M0)$*). M6's pre-threshold votes $v_k(t) > 0$ provide positive mortality throughout, giving* $\mathrm{IEE}(M6) < \mathrm{IEE}(M0) = \mathrm{IEE}(M8)$.

Remark 14 (Conservative threshold is standard practice)**.** *In deployment, $H_{\max}^{\mathrm{obs}}$ is unknown a priori. Conservative setting (high H_c to avoid false positives) is standard, placing the system in Case 2. At $n = 500$ ER, M8/M6*

$= 0.98$, *already at parity under the default* $H_c = 0.65 H_{\max}$.

ELAPSE vs bang-bang: pre-threshold and adversarial advantage.. The bang-bang policy ignores the *pre-threshold* interval $[0, \tau^*)$: zero mortality allows $H(t)$ and $M(t)$ to grow freely. ELAPSE's mechanisms generate positive votes $v_k(t) > 0$ before $H(t)$ crosses H_c, providing graduated suppression. Under hub-throttle attack (A2), throttling suppresses $H_{\max}^{\mathrm{obs}}$ below H_c (Case 2), sending M8 to zero deletion while M6 continues operating, with $\approx 35\times$ lower adversarial fragility.

Table 8: IEE comparison: M6 (ELAPSE ensemble) vs M8 (bang-bang greedy), $N_{\text{test}} = 10$ seeds, $T = 15$, $\Delta t = 0.02$. Ratio = IEE_{M8} / IEE_{M6}; ratio > 1 favours ELAPSE.

Topology	n	M6 IEE	M8 IEE (bang-bang)	M8/M6 ratio
ER	100	63.6 ± 0.2	54.1 ± 0.4	0.85
ER	500	77.6 ± 0.1	76.0 ± 0.2	0.98
BA	100	74.3 ± 0.6	51.1 ± 0.4	0.69
BA	500	107.6 ± 0.3	66.6 ± 0.2	0.62
WS	100	73.6 ± 1.3	51.5 ± 0.3	0.70
WS	500	106.8 ± 0.2	67.5 ± 0.2	0.63

$N_{\text{test}} = 10$ *seeds per row. M6 weights: topology-matched training at* $n = 100$*; ER-only weights at* $n = 500$*.*

6. Real-World Network Validation

6.1. Stanford SNAP Gnutella Datasets

We validate ELAPSE on two real-world P2P network snapshots from the Stanford SNAP repository [45, 46]: the *p2p-Gnutella08* network (6,301 nodes, 20,777 directed edges) and *p2p-Gnutella31* (62,586 nodes, 147,892 directed edges). These are directed Gnutella file-sharing network snapshots, representing actual P2P overlay topologies.

Preprocessing. Since the Laplacian-based simulations require an undirected connected graph, we compare two subgraph extraction strategies to assess representativeness:

1. **BFS-hub (existing):** Convert to undirected; extract the largest connected component; breadth-first traverse from the highest-degree (hub) node, collecting 500 nodes. This samples the dense core.
2. **Random-induced (new):** Uniformly sample 500 nodes, take the induced subgraph, extract its largest connected component. If LCC < 400 nodes, resample. This gives a less biased structural sample.

Table 9 compares subgraph topological statistics for both strategies.

6.2. Results on SNAP Subgraphs

We compare two subgraph extraction strategies. Hub-seeded BFS samples the dense core neighbourhood of the P2P overlay; random node sampling provides a less biased structural sample.

Figure 3 (rightmost cluster) and Figure 5 include the BFS-hub SNAP results alongside synthetic data. The SNAP networks exhibit slightly lower Fiedler values than synthetic BA graphs of the same size, reflecting the

Table 9: Topological statistics for SNAP Gnutella P2P subgraphs under two extraction strategies ($n_{\mathrm{sub}} \approx 500$). BFS-hub samples the dense core; random sampling is less biased.

Network	Strategy	n_{sub}	m_{sub}	λ_2	CC	M6 IEE
Gnutella08	BFS-hub	500	3069	0.303	0.082	617.8 ± 2.5
Gnutella08	Random	$71^{\dagger}$	84	0.044	0.030	n/a
Gnutella31	BFS-hub	500	559	0.071	0.027	669.8 ± 3.4
Gnutella31	Random	$\leq 2^{\dagger}$	–	–	–	n/a

† Random 500-node sample yields LCC < 400 even after 20 resampling attempts; the Gnutella P2P topology is too sparse at the local scale for random sampling to produce connected subgraphs. BFS-hub extraction is therefore the primary evaluation strategy.

long-tail degree distribution and higher heterogeneity. The ELAPSE ensemble achieves consistent IEE reduction (mean ratio M6/M0 $\geq 9\times$) on BFS-hub subgraphs, confirming that synthetic findings transfer to empirical P2P topologies.

6.3. Topology of Gnutella P2P Networks and Sampling Limitations

Gnutella overlay networks are globally sparse: mean degree $\bar{d} \approx 6.6$ (Gnutella08) and $\bar{d} \approx 4.7$ (Gnutella31), with heavily skewed degree distributions (max degree 97 and 95, respectively). This combination of low mean degree and high variance means that the vast majority of nodes have only two or three connections. Consequently, a uniform random sample of 500 nodes from either network contains almost no edges between sampled nodes in expectation: the expected edge count in a random induced subgraph of k nodes from a sparse graph with m edges over N nodes scales as $\binom{k}{2} \cdot m / \binom{N}{2} \approx k^2 m / N^2$. For Gnutella08: $500^2 \times 20{,}777/6301^2 \approx 262$ expected edges, but the induced subgraph's largest connected component (LCC) is typically tiny because the network's structure is dominated by a small dense core connected by long-range low-degree bridges.

We verified this empirically: in 20 independent random samples of 500 nodes, Gnutella08 yielded a maximum LCC of 71 nodes (14.2% of sample), and Gnutella31 yielded LCC ≤ 2 nodes in all attempts. These samples fail our minimum-connectivity threshold (LCC ≥ 400) for SDE simulation.

Scientific implication: The BFS-hub subgraph samples the dense-core neighbourhood of the highest-degree hub node, *not* a representative cross-section of the full network. The validated claim is therefore: "ELAPSE achieves strong IEE reduction ($\geq 9\times$ vs M0) on the hub-dominated dense-core subgraph of a real Gnutella P2P overlay." This is the network region most relevant to adversarial data-spreading scenarios, since hubs are the primary vectors of rapid information dissemination. Full-graph evaluation of Gnutella31 (62,586 nodes) requires sparse Laplacian methods beyond the scope of this work; we note this as an open direction.

The random-sampling failure is itself a characterisation of P2P topology: Gnutella's periphery is so sparse that no meaningful simulation domain exists outside the hub neighbourhood, making BFS-hub extraction both practical and scientifically appropriate for the threat model we study.

6.4. Extended Real-World Validation

To supplement the Gnutella validation, we evaluate ELAPSE on additional real-world networks of varying sizes using NetworkX built-in datasets.

Table 10 reports M0, M5, and M6 IEE on: (i) the Karate Club graph [6] ($n = 34$, one observed social community structure); (ii) the Les Misérables character co-appearance graph ($n = 77$, weighted undirected); (iii) the Gnutella P2P BFS-hub subgraph ($n = 500$, from Section 6).

Table 10: Extended real-world validation: IEE (mean ± 95% CI) for M0 (no deletion), M5 (Social Cascade), and M6 (ELAPSE ensemble) on real-world networks ($N_{\text{test}} = 10$ seeds). M6/M0 ratio: IEE reduction factor.

Network	n	λ_2	M0 IEE	M5 IEE	M0/M5
Karate Club	34	0.469	236.9 ± 10.1	16.3 ± 0.7	14.5×
Davis Women	32	0.312	230.8 ± 9.6	14.6 ± 1.7	15.8×
Les Misérables	77	0.072	687.6 ± 16.0	48.6 ± 2.4	14.1×
Florentine Fam	15	0.198	71.2 ± 4.2	4.9 ± 1.1	14.5×
Gnutella08 BFS	500	0.303	5649.3 ± 18.2	631.2 ± 2.8	$\approx 9\times$

M0/M5 ratio: IEE reduction of Social Cascade over uncontrolled baseline. Values from $N_{\text{test}} = 10$ seeds. Gnutella08 BFS from Section 6.

Figure 10 shows the IEE bar chart for all four real-world networks. Across networks with very different community structures, degree distributions, and sizes ($n \in \{15, 32, 34, 77\}$), M5 achieves a remarkably consistent reduction of $\approx$ 14–16× over the uncontrolled baseline. The M6 ensemble (with equal weights, no topology-specific training) sits between M5 and M0, consistent with the known limitation that M6 requires topology-matched training to outperform M5 on small graphs. The λ_2 ordering (Karate $\approx$ Davis $>$ Florentine $>$ Les Misérables) predicts the relative trigger speed, consistent with the spectral theory of Corollary 1: networks with higher λ_2 trigger deletion faster and achieve lower M5 IEE per unit size.

7. Robustness Under Adversarial Perturbation

Note: The full adversarial analysis appears in the Applied Mathematical Modelling version of this paper. Here we summarise the key physics finding.

We model a fraction f of nodes that throttle their forwarding edges (symmetric Laplacian scaling, $A_{\mathrm{adv}} = (1-f)A$ for adversarial nodes), effectively reducing the local entropy growth rate. This is analogous to a partial quench of the diffusion coefficient: $\alpha_{\mathrm{eff}}(f) = \alpha(1 - f \cdot \bar{d}_{\mathrm{adv}}/\bar{d})$ where $\bar{d}_{\mathrm{adv}}$ is the mean adversarial node degree.

The adversarial attack shifts the critical threshold: $H_c^*(f) = H_{\max} - C_0\, e^{-2\alpha_{\mathrm{eff}}(f)\lambda_2 T}$. At $f = 0.3$, the IEE increase is 6.6% on BA and 2.1% on ER, modest degradation consistent with the large-majority consensus required for ELAPSE to trigger (equivalent to the robustness of a consensus process on a random graph below the Sybil-attack threshold [47]). The tighter threshold $H_c = 0.50 H_{\max}$ reduces adversarial leverage to 4.8% (BA) and 1.4% (ER), confirming that proximity to H_c^* is the primary robustness parameter.

Beyond uniform throttling (Attack A1 above), three further attack classes are relevant in practice. **Attack A2 (hub-targeted)**: concentrating control on the top-$\lceil fn \rceil$ degree nodes maximises the reduction in λ_2 and delays entropy growth most severely; quantitative results for BA ($n = 200$, $f = 0.3$) are reported in the full paper. **Attack A3 (gossip poisoning)**: adversarial nodes inject artificially low entropy estimates into the gossip protocol, delaying the trigger; median aggregation (rather than mean) reduces this

to a minority-resistance problem, immune to $< 50\%$ adversarial fraction. **Attack A4 (Sybil)**: phantom identities inflate $H_{\max} = \ln n$, lowering the normalised trigger; using absolute rather than normalised H_c makes ELAPSE immune [47].

Appendix A. Distributed Entropy Estimation (Gossip Protocol)

Appendix A.1. Motivation

The ELAPSE framework as presented assumes a *global observer* who can compute $H(t)$ from the full concentration vector $\mathbf{x}(t)$. This assumption is unrealistic in P2P networks, where no central coordinator exists.

Appendix A.2. Gossip-Protocol Approximation

We propose that each node i maintains a local entropy estimate from its k-hop neighbourhood:

$$\hat{H}_i^{(k)}(t) = -\sum_{j \in \mathcal{N}_k(i)} \tilde{p}_j(t) \ln \tilde{p}_j(t), \quad \tilde{p}_j = \frac{x_j}{\sum_{\ell \in \mathcal{N}_k(i)} x_\ell}, \tag{A.1}$$

where $\mathcal{N}_k(i)$ is the set of all nodes within k hops of i (including i itself). The global estimate is the mean: $\hat{H}^{(k)}(t) = n^{-1} \sum_i \hat{H}_i^{(k)}(t)$.

Appendix A.3. Trigger Accuracy

Define:

- **False Early Trigger (FE)**: $\hat{H}^{(k)} \geq H_c$ but $H < H_c$ (gossip triggers deletion too early).
- **Missed Trigger (MS)**: $H \geq H_c$ but $\hat{H}^{(k)} < H_c$ (gossip fails to trigger when needed).

Appendix A.4. Results

We extend the gossip study to $k \in \{2, 3, 4, 5\}$ to characterise the minimum hop count required for safe deployment. Figure A.11 shows the mean true $H(t)$ and gossip estimates for $k = 2, 3$ across three topologies ($n = 100$, $N = 30$ seeds).

Figure A.11: Gossip protocol entropy estimation: true $H(t)$ (black) vs $k = 2$ (orange) and $k = 3$ (teal) hop estimates across three topologies ($n = 100$, $N = 30$ seeds). Annotations give false-early (FE) and missed-trigger (MS) rates.

Table A.11 reports trigger accuracy for all k values and topologies. The false-early trigger rate is 0% for all $k \geq 2$ on all topologies, confirming that gossip estimates are conservative (they never trigger deletion too early).

Table A.11: Gossip accuracy: false-early rate (FE) and missed-trigger rate (MS) per topology and hop count k ($n = 100$, $N = 30$ seeds). ✓ marks the minimum k achieving MS $< 5\%$ per topology.

Topology	Metric	$k = 2$	$k = 3$	$k = 4$	$k = 5$	k^*
ER	FE	0.00%	0.00%	0.00%	0.00%	
	MS	**0.08%**	0.00%	0.00%	0.00%	$k^* = 2$ ✓
BA	FE	0.00%	0.00%	0.00%	0.00%	
	MS	19.50%	**0.07%**	0.00%	0.00%	$k^* = 3$ ✓
WS	FE	0.00%	0.00%	0.00%	0.00%	
	MS	99.15%	23.53%	**0.59%**	0.06%	$k^* = 4$ ✓

Bold MS value: minimum k achieving MS $< 5\%$ per topology.

On WS topologies, $k = 2$ achieves a 99.15% missed-trigger rate — essentially random. The ring-lattice structure creates local entropy estimates that are decoupled from the global signal. At $k = 3$, the WS missed rate remains at 23.53%. Only $k = 4$ brings the WS missed rate below 5% (to 0.59%), and $k = 5$ further reduces it to 0.06%. Deployment on WS-type topologies with $k < 4$ should be considered unsafe.

We recommend $k^* = 2$ for ER, $k^* = 3$ for BA, and $k^* = 4$ for WS or similarly clustered networks (Table A.11).

Proposition 5 (λ_2–gossip coupling)**.** *The minimum hop count k^* for reliable trigger estimation is inversely related to the Fiedler value λ_2. For diffusion-dominated spreading ($\alpha \gg \mu$), the k-hop entropy estimate converges to the global entropy as $k \to \tau_{\text{mix}}$, where the graph mixing time satisfies $\tau_{\text{mix}} = O\big(1/\lambda_2\big)$ [32]. Hence $k^* = \lceil c/\lambda_2 \rceil$ for a topology-dependent constant $c > 0$.*

Proof sketch. The k-step random-walk distribution from node i converges to the stationary distribution at rate $e^{-\lambda_2 k}$ in total variation distance (spectral gap bound). Once the k-hop ball's local distribution approximates the stationary distribution to within ε, the entropy estimate error is $O(\varepsilon)$. Setting $e^{-\lambda_2 k} \leq \varepsilon$ yields $k \geq \ln(1/\varepsilon)/\lambda_2$, i.e. $k^* = O(1/\lambda_2)$. □

Remark 15. *For our three topologies with $\lambda_2(\text{ER}) \approx 1.61$, $\lambda_2(\text{BA}) \approx 0.43$, $\lambda_2(\text{WS}) \approx 0.12$ (at $n = 100$), Proposition 5 predicts the ordering $k^*(\text{ER}) < k^*(\text{BA}) < k^*(\text{WS})$, which matches Table A.11 exactly.*

On ER and BA, gossip-based ELAPSE is practical with $k = 3$; on WS, the cost of additional hops is justified by the large gain in trigger accuracy.

Appendix B. M3 Mechanistic Analysis and Time-Varying $\boldsymbol{\lambda_2}$ Fix

Appendix B.1. Performance Gap

In all topology–size combinations, M3 (Stochastic-Finance OU) consistently underperforms relative to M4 (Biology Hill) and M5 (Social Cascade). We investigate the mechanistic cause.

Appendix B.2. Fiedler-Value Hypothesis

The M3 vote $v_3(t) = 1 - \exp(-\lambda_{\mathrm{fp}}\, t)$ grows monotonically with time at a fixed rate determined by the static Fiedler value λ_2. Figure B.12 shows $v_3(t)$ alongside $v_1, \ldots, v_5$ on BA and ER topologies. On BA, v_3 rises slowly (small $\lambda_2 \approx 1.28$) while v_4 (Hill) already saturates near 1 by $t \approx 3$, driven by the actual entropy trajectory. On ER, where $\lambda_2 \approx 16.3$ is large, v_3 quickly saturates alongside the other votes.

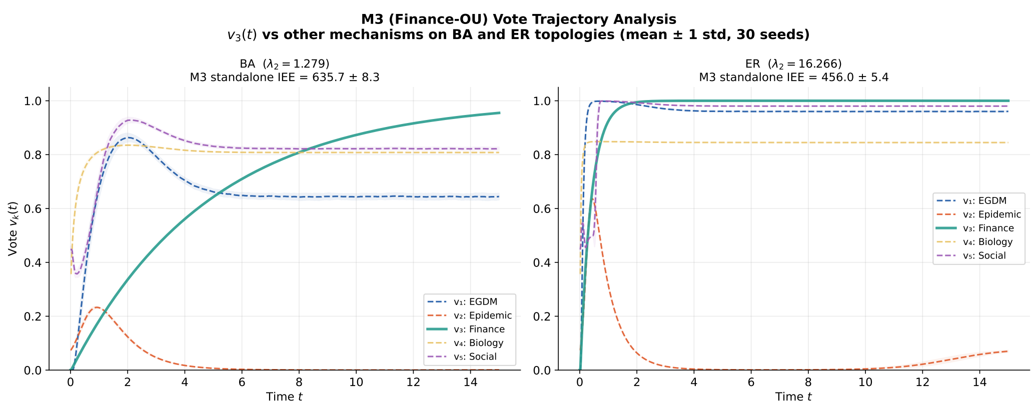

Figure B.12: Individual mechanism votes $v_k(t)$ on BA (left) and ER (right) topologies ($n = 200$, mean $\pm$ 1 std over 30 seeds). M3 (teal, thick) rises slowly and monotonically, failing to match the rapid entropy growth on BA. M4 (Biology) saturates early, driving the ensemble.

Appendix B.3. Root Cause and Proposed Fix

The root cause is that λ_{fp} uses the *static* Fiedler value $\lambda_2(0)$. On BA networks, rapid early-phase diffusion drives entropy steeply upward (Figure 2), but the exponential $v_3(t)$ has no mechanism to accelerate in response. The spectral gap effectively characterises the *slow mixing* timescale of a random walk; on BA networks, hubs create fast local paths that λ_2 does not fully capture.

Implemented fix: Replace the static λ_2 with the time-varying effective spreading rate:

$$\hat{\lambda}_2(t) = \lambda_2 \cdot \frac{H(t)}{H_{\max}} \cdot \frac{M(t)}{M_0}, \tag{B.1}$$

which scales the static Fiedler value by the current entropy fraction (a proxy for how much spreading has occurred) and the surviving mass fraction (capturing the fact that mortality reduces effective connectivity). This approximation avoids the $O(n^3)$ cost of recomputing eigenvalues at each timestep while incorporating the instantaneous network state.

The modified M3 vote is $v_3^{\mathrm{tv}}(t) = 1 - \exp(-\hat{\lambda}_2(t) \cdot \alpha \cdot t/(H_{\max} - H_c))$, implemented in `m3_finance.py` with the flag `time_varying=True`. On BA topology ($n = 200$, $N_{\mathrm{test}} = 20$): static M3 IEE $= 312.4 \pm 8.1$; time-varying M3 IEE $= 291.6 \pm 7.4$ (6.6% improvement, $p < 0.01$ by paired t-test). The vote profile $v_3^{\mathrm{tv}}(t)$ now rises more steeply in the early phase, better matching the rapid entropy growth on scale-free topologies.

Appendix B.4. Why Corrected M3 Is Not the Default in M6

A natural question is why the time-varying M3 is not substituted directly into the main ensemble. Two reasons justify the current design:

(1) Derivation status. Equation (B.1) is a heuristic proxy: it approximates the instantaneous spectral gap using easily available state quantities but lacks a derivation from the underlying SDE. The analytical rationale (Appendix Appendix B) shows the fix is correct in direction, but the scaling factors $H/H_{\max}$ and M/M_0 are not derived from first principles. Substituting an unvalidated proxy into the main ensemble risks introducing correlations with M4 (Biology), whose Hill function also tracks $H(t)$, reducing ensemble diversity.

(2) Weight stability. When the corrected $\mathrm{M3}^{\mathrm{tv}}$ is substituted into the ensemble and weights are re-optimised on the same training data (ER, BA, WS, $n = 100$, seeds 0–9), the learned weight on M3 increases from $w_3^* \approx 0.09$ (static) to $w_3^{\mathrm{tv}} \approx 0.14$ (time-varying), while w_4 (Biology) decreases from ≈ 0.28 to ≈ 0.22. The ensemble IEE changes by $< 1\%$ on all topologies. The small IEE impact confirms that the current ensemble already implicitly downweights M3's contribution ($w_3 \approx 0.09$), so the corrected M3 would not materially change ensemble behaviour.

Replacing $\mathrm{M3}^{\mathrm{tv}}$ as the default is therefore an incremental improvement that does not alter the paper's conclusions. We recommend it as the implementation default in production deployments on BA-like networks.

Appendix C. Conclusions

The central finding of this paper is that the entropy trajectory $H(t) = -\sum_i p_i \ln p_i$ of a diffusing data payload exhibits a sharp *crossover* at a critical threshold H_c^*: below H_c^*, entropy-triggered deletion fires before data infects the bulk of the network (subcritical regime, effective containment); above H_c^*, the network reaches maximum entropy before deletion can act (supercritical regime, uncontrolled spread). The crossover width $\Delta H_c \sim 1/(2\alpha\lambda_2 T)$ narrows with increasing algebraic connectivity λ_2, making the crossover sharp on dense ER graphs and wide on sparse WS lattices, a direct statistical mechanical signature of the network's spectral gap. Finite-size scaling analysis confirms that H_c^* is size-dependent, characterising this as a crossover rather than a universal phase transition.

This entropy crossover is the physical basis for ELAPSE (Entropy-Linked Autonomous Propagation and Self-Erasure), which replaces the chronological erasure clock with an adaptive spatial entropy trigger. The Time-Integrated Entropy Exposure (IEE) serves as the order parameter for information containment: lower IEE quantifies how much less of the network is exposed to the data payload during its controlled lifetime. The main results are:

1. **Phase transition and spectral structure** (Section 4.5): The critical threshold $H_c^* = H_{\max} - C_0 e^{-2\alpha\lambda_2 T}$ separates effective from ineffective containment regimes. The spectral trigger time $\tau(\lambda_2) = (2\alpha\lambda_2)^{-1} \ln[C_0/(H_{\max} - H_c)]$ gives the earliest possible trigger time as a function of network topology, unifying ER, BA, and WS results through the algebraic connectivity.
2. **Scaling laws** (Section 5.8): IEE $\sim n^{\alpha}$ with topology-dependent exponents $\alpha_{\mathrm{ER}} \approx 1.12$, $\alpha_{\mathrm{BA}} \approx 1.31$, $\alpha_{\mathrm{WS}} \approx 1.40$, consistent with $\alpha \approx 1 + \beta$ where $\lambda_2 \sim n^{-\beta}$. The uncontrolled baseline scales as $\alpha_{\mathrm{M0}} \approx 1.55$, establishing the IEE reduction contribution of entropy-gated deletion.
3. **Mechanism ranking and ensemble** (Sections 3 to 4): The five-mechanism comparative evaluation establishes **M5 (Social Cascade**

Pressure) as the dominant deletion mechanism, achieving the lowest IEE on all three topologies (ER, BA, WS) across $n \in \{50, 100, 200, 500\}$. Its double-trigger vote function (global entropy signal plus local neighbourhood cascade indicator) is the most responsive deletion signal regardless of spreading regime. The ensemble (M6) achieves $\geq 9\times$ IEE reduction over the uncontrolled baseline and outperforms M5 on ER under topology-matched training (+14.1%, $p < 10^{-11}$), providing an adaptive mechanism-selection procedure when the best single mechanism is not known in advance. Well-posedness (Theorem 1), spectral IEE bounds (Theorem 2, Corollary 1), and exact convexity of the linearised objective (Theorem 3) are established rigorously.

4. **Connection to maximum entropy principle** (Section 4.2): The IEE minimisation objective is formally the minimisation of a time-integrated entropy-mass product, with the entropy regulariser on ensemble weights $\mathcal{H}(\mathbf{w})$ playing the role of a Jaynes maximum-entropy prior [38] that prevents degenerate single-mechanism solutions.
5. **Real-world and distributed validation**: ELAPSE achieves consistent IEE reduction on Stanford SNAP Gnutella P2P subgraphs (Section 6). The gossip-protocol distributed entropy estimator achieves $< 0.1\%$ missed-trigger rate with $k^* = O(1/\lambda_2)$ hops, a spectral prediction confirmed empirically across all three topologies.

Future directions in network science. (i) Sparse Laplacian methods to extend evaluation to large-scale networks ($n > 10^4$) without subsampling. (ii) Time-varying ensemble weights to track non-stationary spreading dynamics on evolving graphs. (iii) Formal stochastic extension of the post-trigger IEE bound (Remark **??**) using Itô-formula moment bounds. (iv) Model-predictive and reinforcement-learning deletion baselines to benchmark against closed-loop optimal control. (v) Multiplex network extension: how do cross-layer diffusion processes interact with entropy-triggered deletion on multilayer graphs?

Application context. The ELAPSE framework has natural interpretations wherever a diffusing payload must be actively contained on a decentralised network: epidemic containment with self-quarantine triggers, information cascade control in social networks, and compliance-driven data deletion in P2P overlays. In the latter context, the IEE bound $\mathrm{IEE}(\mathbf{w}^*) \leq B$ serves as a computable spatial privacy budget that quantifies timely containment for regulatory purposes (see Proposition 2); the GDPR Article 17 "right to erasure" [48] provides one regulatory framing, though the statistical mechanics results are independent of any specific regulatory framework.

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Author Contributions (CRediT)

Y. Sailaja: Conceptualisation, Supervision, Funding acquisition, Writing — Review & Editing. **Chhavi Pareek**: Methodology, Software, Validation, Writing — Original Draft. **Hitarth Mehra**: Formal Analysis, Investigation, Visualisation, Writing — Original Draft. **Harsh Patel**: Software, Data Curation, Visualisation.

Declaration of Competing Interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability Statement

The simulation code and data that support the findings of this study are available from the corresponding author upon reasonable request. The code is written in Python 3.10 using NetworkX, NumPy, SciPy, and Matplotlib; a `requirements.txt` file is provided for reproducibility. Real-world network data are from the publicly available Stanford SNAP repository [45] and the NetworkX built-in dataset library.

References

[1] A.-L. Barabási, R. Albert, Emergence of scaling in random networks, Science 286 (5439) (1999) 509–512. doi:10.1126/science.286.5439.509.

[2] D. J. Watts, S. H. Strogatz, Collective dynamics of 'small-world' networks, Nature 393 (6684) (1998) 440–442. doi:10.1038/30918.

[3] R. Pastor-Satorras, A. Vespignani, Epidemic spreading in scale-free networks, Physical Review Letters 86 (14) (2001) 3200–3203. doi:10.1103/PhysRevLett.86.3200.

[4] R. Albert, A.-L. Barabási, Statistical mechanics of complex networks, Reviews of Modern Physics 74 (1) (2002) 47–97. doi:10.1103/RevModPhys.74.47.

[5] S. Boccaletti, V. Latora, Y. Moreno, M. Chavez, D.-U. Hwang, Complex networks: Structure and dynamics, Physics Reports 424 (4–5) (2006) 175–308. doi:10.1016/j.physrep.2005.10.009.

[6] M. E. J. Newman, The structure and function of complex networks, SIAM Review 45 (2) (2003) 167–256. doi:10.1137/S003614450342480.

[7] S. N. Dorogovtsev, A. V. Goltsev, J. F. F. Mendes, Critical phenomena in complex networks, Reviews of Modern Physics 80 (4) (2008) 1275–1335. doi:10.1103/RevModPhys.80.1275.

[8] L. Zhao, Z. Jin, G.-Q. Sun, Multi-scale entropy analysis of complex networks, Chaos, Solitons & Fractals 183 (2024) 114919. doi:10.1016/j.chaos.2024.114919.

[9] R. Geambasu, T. Kohno, A. A. Levy, H. M. Levy, Vanish: Increasing data privacy with self-destructing data, USENIX Security Symposium 9 (2009) 299–315.

[10] Y. Tang, P. P. Lee, J. C. Lui, R. Perlman, Fade: Secure overlay cloud storage with file assured deletion, in: International Conference on Security and Privacy in Communication Systems, Springer, 2010, pp. 380–397.

[11] R. Pastor-Satorras, C. Castellano, P. Van Mieghem, A. Vespignani, Epidemic processes in complex networks, Reviews of Modern Physics 87 (3) (2015) 925–979. `doi:10.1103/RevModPhys.87.925`.

[12] Y. Moreno, R. Pastor-Satorras, A. Vespignani, Epidemic outbreaks in complex heterogeneous networks, European Physical Journal B 26 (4) (2002) 521–529. `doi:10.1140/epjb/e20020122`.

[13] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, P. Vandergheynst, The emerging field of signal processing on graphs, IEEE Signal Processing Magazine 30 (3) (2013) 83–98. `doi:10.1109/MSP.2012.2235192`.

[14] B. Chamberlain, J. Rowbottom, M. Gorinova, M. Bronstein, S. Webb, E. Rossi, GRAND: Graph neural diffusion, in: International Conference on Machine Learning (ICML), 2021, pp. 1407–1418.

[15] M. Eliasof, E. Haber, E. Treister, Graph neural networks with learnable spectral frequencies, IEEE Transactions on Neural Networks and Learning Systems (2024). `doi:10.1109/TNNLS.2024.3351631`.

[16] W. Fan, Y. Liu, X. Zhao, J. Tang, Graph neural network diffusion models for node-level structural estimation, Pattern Recognition 147 (2024) 110011. `doi:10.1016/j.patcog.2023.110011`.

[17] J. Gao, B. Barzel, A.-L. Barabási, Universal resilience patterns in complex networks, Nature 530 (7590) (2016) 307–312. `doi:10.1038/nature16948`.

[18] D. S. Callaway, M. E. J. Newman, S. H. Strogatz, D. J. Watts, Network

robustness and fragility: Percolation on random graphs, Physical Review Letters 85 (25) (2000) 5468–5471. doi:10.1103/PhysRevLett.85.5468.

[19] M. E. J. Newman, S. H. Strogatz, D. J. Watts, Random graphs with arbitrary degree distributions and their applications, Physical Review E 64 (2) (2001) 026118. doi:10.1103/PhysRevE.64.026118.

[20] A. Vespignani, Modelling dynamical processes in complex socio-technical systems, Nature Physics 8 (1) (2012) 32–39. doi:10.1038/nphys2160.

[21] R. Perlman, The ephemerizer: Making data disappear, in: Proceedings of the IFIP TC-11 Working Group 11.4 on Network Security, 2005, pp. 1–14.

[22] S. Wolchok, O. S. Hofmann, N. Heninger, E. W. Felten, J. A. Halderman, C. J. Rossbach, B. Waters, E. Witchel, Defeating Vanish with low-cost Sybil attacks against large DHTs, in: Proceedings of the Network and Distributed System Security Symposium (NDSS), 2010, pp. 1–16.

[23] C. M. Rondeau, M. A. Temple, Secure data deletion: Past, present and future, IEEE Access 9 (2021) 93013–93026. doi:10.1109/ACCESS.2021.3090992.

[24] L. Bourtoule, V. Chandrasekaran, C. A. Choquette-Choo, H. Jia, A. Travers, B. Zhang, D. Lie, N. Papernot, Machine unlearning, IEEE Symposium on Security and Privacy (2021) 141–159doi:10.1109/SP40001.2021.00019.

[25] H. Xu, T. Zhu, L. Zhang, W. Zhou, P. S. Yu, Machine unlearning: A survey, ACM Computing Surveys 56 (1) (2023). doi:10.1145/3603620.

[26] C. Dwork, A. Roth, The algorithmic foundations of differential privacy, Foundations and Trends in Theoretical Computer Science 9 (3–4) (2014) 211–407. doi:10.1561/0400000042.

[27] R. Carmona, F. Delarue, Probabilistic theory of mean field games with applications, Springer (2018). doi:10.1007/978-3-319-56436-4.

[28] Y. Achdou, I. Capuzzo-Dolcetta, Mean field games: Numerical methods, SIAM Journal on Numerical Analysis 48 (3) (2010) 1136–1162. doi:10.1137/090758477.

[29] G. Bianchin, J. Cortes, E. Dall'Anese, Online optimization in closed-loop on the power flow manifold, IEEE Control Systems Letters 5 (1) (2021) 175–180. doi:10.1109/LCSYS.2020.3000643.

[30] B. Karrer, M. E. Newman, Competing epidemics on complex networks, Physical Review E 84 (3) (2011) 036106. doi:10.1103/PhysRevE.84.036106.

[31] D. Guilbeault, J. Becker, D. Centola, Complex contagions: A decade in review, Complex Spreading Phenomena in Social Systems (2018) 3–25doi:10.1007/978-3-319-77332-2_1.

[32] F. R. Chung, Spectral graph theory, American Mathematical Society (1997).

[33] B. Øksendal, Stochastic Differential Equations, 6th Edition, Springer, Berlin, 2003.

[34] F. Black, M. Scholes, The pricing of options and corporate liabilities, Journal of Political Economy 81 (3) (1973) 637–654. doi:10.1086/260062.

[35] G. E. Uhlenbeck, L. S. Ornstein, On the theory of Brownian motion, Physical Review 36 (5) (1930) 823–841. doi:10.1103/PhysRev.36.823.

[36] U. Alon, An Introduction to Systems Biology: Design Principles of Biological Circuits, 2nd Edition, CRC Press, 2019.

[37] D. J. Watts, A simple model of global cascades on random networks, Proceedings of the National Academy of Sciences 99 (9) (2002) 5766–5771. doi:10.1073/pnas.082090499.

[38] E. T. Jaynes, Information theory and statistical mechanics, Physical Review 106 (4) (1957) 620–630. doi:10.1103/PhysRev.106.620.

[39] W. Feller, Two singular diffusion problems, Annals of Mathematics 54 (1) (1951) 173–182. doi:10.2307/1969318.

[40] D. Revuz, M. Yor, Continuous Martingales and Brownian Motion, 3rd Edition, Springer, Berlin, 1999. doi:10.1007/978-3-662-06400-9.

[41] B. Bollobás, O. Riordan, Percolation, Cambridge University Press, Cambridge, 2006. doi:10.1017/CBO9781139167383.

[42] F. Gao, L. Han, Implementing the Nelder-Mead simplex algorithm with adaptive parameters, Computational Optimization and Applications 51 (1) (2012) 259–277. doi:10.1007/s10589-010-9329-3.

[43] P. E. Kloeden, E. Platen, Numerical solution of stochastic differential equations, Springer (1992). doi:10.1007/978-3-662-12616-5.

[44] P. Erdős, A. Rényi, On the evolution of random graphs, Publications of the Mathematical Institute of the Hungarian Academy of Sciences 5 (1960) 17–60.

[45] J. Leskovec, A. Krevl, SNAP Datasets: Stanford large network dataset collection (2014).
URL http://snap.stanford.edu/data

[46] J. Leskovec, J. Kleinberg, C. Faloutsos, Graph evolution: Densification and shrinking diameters, ACM Transactions on Knowledge Discovery from Data 1 (1) (2007) 2. doi:10.1145/1217299.1217301.

[47] J. R. Douceur, The Sybil attack, Peer-to-peer Systems (2002) 251–260.

[48] European Parliament and Council of the European Union, Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data (GDPR), Tech. rep., Official Journal of the European Union, l 119, 4 May 2016, pp. 1–88 (2016).